CONTINUOUS AND SMOOTH PLURISUBHARMONIC EXHAUSTIONS: THREEFOLD COUNTEREXAMPLES AND SURFACE REGULARIZATION

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Abstract. Starting from Koike's modified Grauert example, we construct a smooth quasi-projective Kähler threefold with a locally Lipschitz plurisubharmonic exhaustion but no C^2 plurisubharmonic exhaustion. A modification over one point gives a non-Kähler threefold with the same properties. In dimension two, we prove that every divisor complement in a compact Kähler surface admitting a continuous plurisubharmonic exhaustion admits a smooth one. In particular, this holds for every smooth complex quasi-projective surface. The proof averages positive dd^c-closed currents over exhaustion levels to eliminate their Lelong numbers, then combines the surface current theorem of Dinh–Nguyên–Sibony with the Hodge index theorem and the intersection matrix of the boundary. In the non-Stein-modification case, a positive integral divisor supported on the entire boundary has a flat Hermitian metric near the boundary and yields a pluriharmonic logarithmic collar. We retain numerical existence criteria for nonprojective compactifiers and smooth rulings, and sufficient conditions from Ueda theory. The compactification hypotheses and the recent elliptic-neighborhood input to these separate existence results are explicit. In arbitrary dimension, we also show that a reduced pure n-dimensional complex space with a continuous psh exhaustion and a holomorphic map to $\mathbb{C}^{n-1}$ whose fibers all have dimension at most one is holomorphically convex, hence admits a smooth psh exhaustion, without any restriction on its singular locus. This criterion combines geometric flatness and cycle compactness with Vîjîtu's convexity criteria. The unrestricted regularization problem for noncompact complex surfaces is not resolved here.

1. Introduction and main results

The regularity of a plurisubharmonic exhaustion matters even for Kähler manifolds. This paper constructs a smooth quasi-projective threefold admitting a continuous plurisubharmonic exhaustion but no C^2 one, and then inserts a non-Kähler configuration over a single point without changing

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these exhaustion properties. In contrast, we prove continuous-to-smooth regularization for divisor complements in compact Kähler surfaces, including every smooth quasi-projective surface. A further affirmative result in every dimension uses holomorphic maps with one-dimensional fibers and also permits a singular total space, with no restriction on its singular locus. Throughout the introduction, psh abbreviates plurisubharmonic, and smooth means C^∞.

1.1. **Threefold counterexamples and their metric construction.** Our compact base is the modified Grauert ruled surface appearing in Koike's work [1, Example 4.1]. Its failure of semipositivity is already known; the first-order obstruction, including its C^2 version, is not claimed as a new result here [1, Theorem 1.4 and Example 4.1], [2, Remark 2.8]. We include a direct proof to make the regularity threshold used below explicit. The analytic construction then consists of a cusp correction, a global maximum, and a radius criterion converting any hypothetical C^2 exhaustion into a forbidden metric. Failure of smooth semipositivity by itself would not provide a continuous exhaustion.

The cusp estimate also reaches the locally Lipschitz endpoint: a sufficiently large multiple of $q^{1/2}$ dominates the initial curvature in the scaled tangent variables. The coefficient is essential at this endpoint; for smaller exponents shrinking the neighborhood already suffices.

Theorem 1.1. *Let C be the smooth projective completion of*

$$y^2 = x^5 - x,$$

let $K = K_C$, and let

$$0 \longrightarrow K \longrightarrow V \longrightarrow \mathcal{O}_C \longrightarrow 0 \tag{1.1}$$

be a nonsplit holomorphic extension. Let

$$S = \mathbf{P}_{\text{lines}}(V), \qquad \pi : S \longrightarrow C, \qquad Y = \mathbf{P}(K) \subset S,$$

and set

$$L = \mathcal{O}_S(Y) \otimes \pi^* K, \qquad M = \operatorname{Tot}(L^{-1}). \tag{1.2}$$

Then:

- (i) *S is a smooth projective surface and M is a smooth quasi-projective Kähler threefold;*
- (ii) *L admits a continuous Hermitian metric with plurisubharmonic local weights, but admits no C^2 Hermitian metric with semipositive curvature;*
- (iii) *M admits a continuous plurisubharmonic exhaustion, but admits no C^2 plurisubharmonic exhaustion.*

The metric and the exhaustion in (ii)–(iii) *can be chosen locally Lipschitz, and hence locally of class $C^{0,\gamma}$ for every $0 < \gamma \leq 1$. No assertion about existence of a C^1 exhaustion is made.*

The passage from metrics to exhaustions uses a criterion valid for a holomorphic line bundle A over any compact complex manifold Z:

$$\operatorname{Tot}(A^{-1}) \text{ admits a } C^r \text{ psh exhaustion}$$
$$\Longleftrightarrow$$
$$A \text{ admits a } C^r \text{ semipositive Hermitian metric}, \qquad r \in \{2, \infty\}.$$

This is Corollary 3.4. A semipositive metric gives the squared dual norm exhaustion. Conversely, averaging an arbitrary exhaustion under fiberwise circle rotation and taking a sufficiently high level yields a radius function whose negative logarithm is a psh metric weight. Consequently, the obstruction in Theorem 1.1 rules out every C^2 psh exhaustion, not only a smoothing of the particular one constructed here.

The non-Kähler example has the same dimension and agrees with M away from a compact exceptional fiber.

Theorem 1.2. *For every point x of the threefold M in Theorem 1.1, there is a smooth non-Kähler threefold N and a proper holomorphic modification $\mu : N \to M$ inducing a biholomorphism*

$$N \setminus \mu^{-1}(x) \simeq M \setminus \{x\}.$$

The threefold N admits a locally Lipschitz psh exhaustion but no C^2 psh exhaustion.

The construction places Hironaka's opposite-order blowups inside the exceptional plane of the blowup of x. It produces two compact rational curves whose sum is zero in real homology, precluding a Kähler form. The exhaustion assertions follow from Proposition 4.1: a proper surjective holomorphic map that is biholomorphic away from a compact subset preserves the existence of C^r psh exhaustions, for $r = 0, 2, \infty$. The reverse implication uses convex composition to make an exhaustion constant near the exceptional set before descending it. Thus the modification is localized over one point; no small deformation of a fixed complex structure is asserted. Choosing x outside the zero section also preserves a neighborhood of the compact base. The compact product principle in Proposition 4.4 gives further examples in higher dimensions, including the Hopf-surface product in Corollary 4.5.

1.2. **Regularization on surfaces with a divisor compactification.** The surface result has a different conclusion from the threefold construction.

Theorem 1.3. *Let S be a smooth compact connected Kähler surface and let $D \neq 0$ be an effective divisor. If $X = S \setminus \operatorname{supp} D$ admits a continuous psh exhaustion, then it admits a smooth psh exhaustion. In particular, every smooth complex quasi-projective surface admitting a continuous psh exhaustion admits a smooth one.*

Theorem 5.4 and Corollary 5.7 prove this statement without assuming that S is minimal or that the boundary is connected, smooth, or has normal crossings. The proof first uses the continuous minimal-kernel criterion of

Mongodi–Słodkowski [5, Lemma 2.1]. Unless X is a modification of a Stein space, sufficiently high exhaustion levels support positive dd^c-closed currents. Finite averages over equally spaced levels have a mass-one limit supported compactly inside X. A uniform Lelong–Jensen estimate and continuity of the exhaustion force all Lelong numbers of this limit to vanish. The surface current theorem of Dinh–Nguyên–Sibony [18, Corollary 2.4 and its proof] then combines with the Hodge index theorem and the boundary intersection matrix to produce a positive integral divisor A with full boundary support, $A^2 = 0$, and a smooth flat Hermitian metric on $\mathcal{O}_S(A)$ near the boundary. Its canonical section yields a smooth pluriharmonic logarithmic collar.

In particular, a continuous divergent psh collar around a smooth square-zero curve in a compact Kähler surface forces a smooth exhaustion, with no restriction on the genus or Ueda type (Corollary 5.9). The proof does not infer convergence merely from infinite Ueda type. Flatness is obtained in the non-Stein-modification alternative by the current and intersection argument.

Section 5 also retains results that establish existence without assuming a continuous exhaustion: numerical boundary criteria for nonprojective compactifiers and smooth rulings, and sufficient conditions from Ueda theory. These are logically distinct from regularization. For arbitrary noncompact surfaces with a nonconstant holomorphic function, Vîjîitu's holomorphic-convexity theorem [4] gives the familiar regularization consequence recorded in Proposition 5.1. Quasi-projectivity does not ensure the existence of such a function, even under the exhaustion hypothesis; see Remark 5.8. For a line bundle over a compact curve, the degree criterion in Proposition 5.2 gives an elementary explanation of why the threefold construction cannot be repeated over a curve.

Table 1 records the resulting classification of the *compactifiers* under a continuous exhaustion hypothesis. Every projective row is affirmative, including nonminimal projective compactifiers. An arbitrary noncompact Kähler surface need not admit a divisor compactification, and an arbitrary analytic domain in a projective surface need not have a divisorial boundary. Neither class is covered by the compactification hypothesis. Sections 2–4 contain the threefold constructions; Section 5 establishes the independent surface results; Section 6 proves the curve-fibering criterion described next.

1.3. **Holomorphic convexity from curve fiberings.** An additional regularization mechanism requires no compactification and works in every dimension.

Theorem 1.4. *Let X be a reduced complex space of pure dimension $n \geq 2$ with a continuous real-valued psh exhaustion. If a holomorphic map*

$$F : X \longrightarrow \mathbb{C}^{n-1}$$

has every nonempty fiber of dimension at most one, then X is holomorphically convex and admits a smooth psh exhaustion. No condition is imposed on the connected components of $\operatorname{Sing} X$.

Theorem 6.1 proves this assertion. The dimension bound concerns every fiber, including exceptional fibers; a generic rank condition would not suffice for the argument. Normality or Cohen–Macaulayness of X, properness of F, and smoothness or reduced scheme structure of its fibers are not assumed. The smooth exhaustion is obtained by pulling back one from the Stein target of the Remmert reduction. Smoothness on a complex space means local restriction of smooth ambient functions; the conclusion is existence of a smooth exhaustion, not approximation of the given one.

The key step is the stability of noncompact irreducible fiber components meeting a fixed compact set (Lemma 6.4). Geometric flatness over the normal base supplies weighted fiber cycles even for a singular source. The psh exhaustion confines compact components to a common compact set, and Bishop compactness then controls their limits. This replaces the smooth-source stability result in [20, Corollary 4.2]; the boundary-collar and sublevel criteria in that paper already apply to complex spaces. Corollary 6.7 identifies the resulting simultaneous contraction of compact fiber chains.

1.4. **Relation to prior work and scope of the claims.** The known ruled-surface obstruction, Hironaka's opposite-order blowups, and the quoted classification and Ueda theorems are inputs, not new constructions of this paper. The self-contained threefold argument combines the explicit bounded-potential metric, the radius obstruction, and a compactly supported modification. The bounded-potential remark at the end of Section 2 explains which part of the question in [3, Section 4] this construction addresses.

The compactifiable Kähler surface theorem combines the published continuous minimal-kernel criterion and the general positive-current theorem with the level-averaging estimate and the full-boundary intersection argument given in Section 5. It uses no elliptic neighborhood theorem, surface abundance theorem, or minimal-model reduction. Within smooth quasi-projective manifolds, the examples in Theorem 1.1 occur in the smallest possible dimension: surfaces are covered by Theorem 1.3, while a connected noncompact Riemann surface is Stein. We do not assert minimality of dimension three among arbitrary noncompact Kähler manifolds, or optimality of Lipschitz regularity as a threshold.

The elliptic-neighborhood step in the separate existence results invokes [14, Theorem 1.6], a September 2026 preprint. That input is independent of both threefold counterexamples and of Theorem 1.3. No classification or general regularization theorem for arbitrary noncompact complex surfaces is asserted.

The curve-fibering criterion is a deduction from classical geometric flatness and cycle compactness together with [20, Corollary 2.5 and Propositions 2.2, 2.4]. Its singular-source formulation does not require the compactness condition on the singular locus in [20, Theorem 1.1]. For an irreducible surface, its convexity conclusion already follows from [4, Theorem 1.1]. We attribute

the underlying convexity and cycle-theoretic inputs explicitly and do not claim priority for this formulation.

1.5. **Conventions and terminology.** We use

$$d^c = \frac{i}{4\pi}(\bar\partial - \partial), \qquad dd^c = \frac{i}{2\pi}\partial\bar\partial.$$

For a holomorphic line bundle A and a local holomorphic frame e, a Hermitian metric is written

$$\|e\|_h^2 = e^{-\varphi}.$$

Its curvature is $\Theta_h(A) = dd^c\varphi$, representing $c_1(A)$. If $e_j = g_{ij}e_i$, the metric weights satisfy

$$\varphi_j = \varphi_i - \log|g_{ij}|^2. \tag{1.3}$$

A positive continuous Hermitian metric is semipositive in the current sense when its local weights are continuous psh functions. For a C^2 metric this agrees with pointwise semipositivity of its curvature.

For a C^2 real-valued function f, its unnormalized Levi form is

$$\mathcal{L}_f(\xi) = \sum_{j,k} f_{z_j\bar z_k}\xi_j\overline{\xi_k}.$$

Thus f is psh exactly when $\mathcal{L}_f \geq 0$. The fixed positive factor $1/(2\pi)$ in dd^c does not affect any positivity argument. Poincaré–Lelong takes the form

$$dd^c \log\|s_D\|_h^2 = [D] - \Theta_h(\mathcal{O}(D)), \tag{1.4}$$

where s_D is the canonical section of an effective divisor D.

Definition 1.5. A continuous real-valued function F on a complex manifold X is an *exhaustion* if every sublevel set $\{F \leq c\}$ is compact. Equivalently, $F(x_j) \to +\infty$ for every sequence escaping every compact subset of X.

We say that X is *continuously weakly complete* or *smoothly weakly complete* when it admits a continuous or smooth psh exhaustion, respectively. Specifying the regularity avoids the varying conventions for weak completeness in the literature. In Section 6 the same terminology is used for reduced complex spaces, with smooth functions understood through local embeddings. Our projectivization $\mathbf{P}_{\text{lines}}(E)$ parametrizes one-dimensional subspaces of the fibers of E, and its tautological line bundle is $\mathcal{O}_{\mathbf{P}(E)}(-1)$.

2. The ruled surface and continuous semipositive metrics

2.1. **The ruled surface and the metric obstruction.**

2.1.1. *The curve and the extension.* The polynomial $x^5 - x$ has five distinct roots. The double covering $C \to \mathbf{P}^1$ is therefore branched at these five roots and at infinity. Riemann–Hurwitz gives

$$2g(C) - 2 = 2(-2) + 6 = 2,$$

so $g(C) = 2$ and $\deg K = 2$. Serre duality gives

$$\operatorname{Ext}^1(\mathcal{O}_C, K) = H^1(C, K) \simeq H^0(C, \mathcal{O}_C)^* \simeq \mathbb{C}.$$

Consequently a nonzero extension class exists. All nonzero classes give isomorphic extensions after allowing scalar automorphisms of the end terms, so the isomorphism class of V is unambiguous.

One can also specify the extension through a Dolbeault operator. Choose a smooth positive $(1,1)$-form β on C with $\int_C \beta = 1$, and regard it as a K-valued $(0,1)$-form. On the smooth bundle $K \oplus \mathcal{O}_C$, set

$$\bar\partial_V(s, f) = (\bar\partial_K s + f\beta, \bar\partial f). \tag{2.1}$$

This operator is integrable, since there are no $(0,2)$-forms on a curve, and defines (1.1). A splitting would give a holomorphic lift $(s, 1)$ and hence $\bar\partial_K s = -\beta$. Since s is a smooth $(1,0)$-form, Stokes' theorem would then imply $\int_C \beta = 0$, a contradiction. The choice of the standard identification of K-valued $(0,1)$-forms with $(1,1)$-forms only changes this argument by a sign.

2.1.2. *The negative section and the line bundle.* By GAGA, V is algebraic, so $S = \mathbf{P}_{\text{lines}}(V)$ is a smooth projective surface. The inclusion $K \hookrightarrow V$ gives a section $Y = \mathbf{P}(K)$ of π. Its normal bundle is

$$N_{Y/S} \simeq \operatorname{Hom}(K, V/K) \simeq K^{-1}. \tag{2.2}$$

In particular,

$$\mathcal{O}_S(Y)|_Y \simeq K^{-1}, \qquad L|_Y \simeq \mathcal{O}_Y, \qquad Y^2 = -2. \tag{2.3}$$

The trivialization of $L|_Y$ in (2.3) is the canonical one induced by $K^{-1} \otimes K$.

For context, this line bundle is nef and big. If F is a fiber of π, then

$$L \cdot Y = 0, \qquad L \cdot F = 1, \qquad L^2 = 2.$$

For every irreducible curve $D \neq Y$ dominating C,

$$L \cdot D = Y \cdot D + 2\deg(\pi|_D) > 0.$$

Every vertical irreducible curve is a fiber. Hence L is nef, and the usual surface criterion $L^2 > 0$ implies bigness. For completeness, the latter also follows from Riemann–Roch: $h^2(S, L^m) = 0$ for m large, because $(K_S - mL)$ has negative intersection with a fixed ample divisor, and therefore

$$h^0(S, L^m) \geq \chi(S, L^m) = m^2 + O(m).$$

Moreover, $2L - Y = Y + 2\pi^* K$ is ample by Nakai–Moishezon: its square is 6, its intersections with Y and a fiber are 2 and 1, and its intersection with every other irreducible curve is positive.

2.1.3. *First-order transition functions.* Choose a finite coordinate cover $\{U_i\}$ of C, frames k_i of K, and holomorphic lifts v_i of 1 in (1.1). On overlaps,

$$k_j = g_{ij}k_i, \qquad v_j = v_i + c_{ij}k_i. \tag{2.4}$$

The cocycle $\{c_{ij}k_i\}$ represents the nonzero extension class. Near Y, write a line in V as

$$[k_i + w_i v_i].$$

Expressing this line in the j-th frame yields

$$w_j = \frac{g_{ij}w_i}{1 - c_{ij}w_i}. \tag{2.5}$$

Shrinking the neighborhoods if necessary, all denominators are nonzero. Let d_i be the frame of $\mathcal{O}_S(Y)$ defined by $s_Y = w_i d_i$. Then

$$\frac{d_j}{d_i} = \frac{w_i}{w_j} = \frac{1 - c_{ij}w_i}{g_{ij}}.$$

Thus the frames $\ell_i = d_i \otimes \pi^* k_i$ of L satisfy

$$\ell_j = (1 - c_{ij}w_i)\ell_i. \tag{2.6}$$

Their restrictions to Y agree, in accordance with the trivialization in (2.3).

Proposition 2.1. *The line bundle L admits no C^2 Hermitian metric with semipositive curvature.*

Proof. Suppose such a metric exists, with C^2 psh weights φ_i in the frames ℓ_i. By (2.6), the functions $\varphi_i|_Y$ agree on overlaps. They define a subharmonic function on the compact connected curve Y, hence a constant. Rescaling the metric, we may assume

$$\varphi_i(z, 0) = 0. \tag{2.7}$$

In particular $\varphi_{i,z\bar{z}}(z, 0) = 0$. The Levi matrix at $(z, 0)$ is positive semidefinite. A positive semidefinite Hermitian matrix with a zero diagonal entry has zero entries in the corresponding row and column. Therefore

$$\varphi_{i,w\bar{z}}(z, 0) = 0.$$

The C^1 function

$$a_i(z) = \varphi_{i,w}(z, 0)$$

satisfies the Cauchy–Riemann equation and is holomorphic.

Equations (1.3) and (2.6) give

$$\varphi_j(z, w_j) = \varphi_i(z, w_i) - \log|1 - c_{ij}(z)w_i|^2,$$

where the base coordinate change is understood. Differentiating with respect to w_i at zero and using (2.5), we obtain

$$g_{ij}a_j = a_i + c_{ij}. \tag{2.8}$$

The holomorphic lifts $v_i' = v_i - a_i k_i$ consequently agree on overlaps:

$$v_j' = v_i + c_{ij}k_i - a_j g_{ij}k_i = v_i - a_i k_i = v_i'.$$

They define a global holomorphic lift of 1, splitting (1.1). This is a contradiction. □

This calculation is the specialized first-order obstruction occurring in [1, Theorem 1.4 and Example 4.1]. Related obstruction classes, including their applicability to C^2 metrics, are discussed in [2]. Neither reference is needed for the preceding proof.

2.2. **A local continuous metric near a negative curve.** We isolate the analytic estimate used in the construction. The unitary flatness assumption means that a smooth metric of zero curvature exists on the restricted line bundle.

Lemma 2.2 (Cusp correction)**.** *Let D be a compact connected smooth curve in a complex surface X. Suppose $\deg N_{D/X} < 0$, and let A be a holomorphic line bundle such that $A|_D$ is unitary flat. Choose a smooth metric h_0 on A restricting to a flat metric on D, and a smooth metric h_D on $\mathcal{O}_X(D)$ whose curvature is strictly negative on TD. Put*

$$q = \|s_D\|^2_{h_D}.$$

For every $0 < a \le \frac{1}{2}$, there is a constant $\Lambda > 0$ such that the local weights of h_0 plus Λq^a are strictly psh off D on a sufficiently small neighborhood of D and extend as continuous psh weights across D. They are locally of class $C^{0,2a}$; in particular they are locally Lipschitz when $a = \frac{1}{2}$. For $0 < a < \frac{1}{2}$ one may take $\Lambda = 1$ after shrinking the neighborhood. At $a = \frac{1}{2}$, a sufficiently large Λ suffices.

Proof. The chosen metrics exist. Negative degree on the curve gives a negative curvature metric on $N_{D/X} = \mathcal{O}_X(D)|_D$. Any metric on a restricted line bundle extends smoothly to a neighborhood: extend the logarithm of its ratio to a fixed smooth metric. The same argument extends the flat metric on $A|_D$.

Use finitely many relatively compact coordinates (z, w) near D, with $D = \{w = 0\}$. Write

$$q = |w|^2 e^{-b(z,w)}, \qquad \beta = -b,$$

where b is the smooth weight of h_D. Negativity along TD says

$$\beta_{z\bar{z}}(z, 0) > 0. \tag{2.9}$$

For $w \neq 0$, set $\tau = \log q = \log |w|^2 + \beta$. The chain rule gives the exact identity

$$\mathcal{L}_{q^a}(v) = q^a(a\mathcal{L}_\tau(v) + a^2|\partial\tau(v)|^2). \tag{2.10}$$

Evaluate this on

$$v = \xi\frac{\partial}{\partial z} + w\nu\frac{\partial}{\partial w}.$$

Since $\log |w|^2$ is pluriharmonic off D,

$$\mathcal{L}_\tau(v) = \beta_{z\bar{z}}(z, 0)|\xi|^2 + O(|w|)(|\xi|^2 + |\nu|^2),$$

and

$$\partial\tau(v) = \nu + \beta_z(z,0)\xi + O(|w|)(|\xi| + |\nu|).$$

Therefore

$$\begin{aligned}\frac{\mathcal{L}_{q^a}(\xi, w\nu)}{q^a} &= a\beta_{z\bar{z}}(z,0)|\xi|^2 + a^2|\nu + \beta_z(z,0)\xi|^2 \\ &\quad + O(|w|)(|\xi|^2 + |\nu|^2). \end{aligned} \tag{2.11}$$

All remainder bounds are uniform after choosing the finite cover and shrinking its coordinate neighborhoods.

The leading Hermitian form in (2.11) is positive definite: vanishing of its first term forces $\xi = 0$ by (2.9), and vanishing of the second then forces $\nu = 0$. Its smallest eigenvalue has a positive uniform lower bound on the compact coordinate pieces covering D. Also q^a is uniformly comparable to $|w|^{2a}$. Thus, after shrinking the neighborhoods,

$$\mathcal{L}_{q^a}(\xi, w\nu) \geq c|w|^{2a}(|\xi|^2 + |\nu|^2) \tag{2.12}$$

for a uniform $c > 0$.

Let φ_0 be a local weight of h_0. Its restriction to D has zero curvature, so $\varphi_{0,z\bar{z}}(z,0) = 0$. Smoothness gives

$$\varphi_{0,z\bar{z}} = O(|w|), \qquad \varphi_{0,z\bar{w}} = O(1), \qquad \varphi_{0,w\bar{w}} = O(1).$$

Consequently, in the same scaled variables,

$$|\mathcal{L}_{\varphi_0}(\xi, w\nu)| \leq C|w|(|\xi|^2 + |\nu|^2). \tag{2.13}$$

Combining (2.12) and (2.13) yields

$$\mathcal{L}_{\varphi_0 + \Lambda q^a}(\xi, w\nu) \geq (\Lambda c|w|^{2a} - C|w|)(|\xi|^2 + |\nu|^2). \tag{2.14}$$

If $0 < a < \frac{1}{2}$, take $\Lambda = 1$ and then shrink until the coefficient is positive. If $a = \frac{1}{2}$, fix $\Lambda > C/c$; the coefficient becomes $(\Lambda c - C)|w| > 0$ throughout the punctured neighborhood where the two uniform estimates hold. The constants c, C were obtained before choosing Λ, so this choice is not circular. Every tangent vector at $w \neq 0$ can be written in these scaled variables. This proves strict plurisubharmonicity off D in both cases.

The function $q^a = |w|^{2a}e^{-ab}$ is continuous and locally $C^{0,2a}$, including $2a = 1$. The local weights $\varphi_0 + \Lambda q^a$ are bounded near D and extend continuously to it. The removable singularity theorem for psh functions across an analytic hypersurface therefore proves that these extended weights are psh. Adding the globally defined function Λq^a preserves the metric transition law (1.3). □

Remark 2.3. The scale in (2.11) is essential. The normal component of the Levi form has order $|w|^{2a-2}$, while the tangential component has order $|w|^{2a}$. These two scales allow the cusp to dominate the bounded mixed curvature of a smooth initial metric. At $a = \frac{1}{2}$ the scaled cusp and the initial error both have order $|w|$, which explains the coefficient choice in (2.14).

Ordinary smoothing of the cusp would not preserve this domination at D. The argument proves no C^1 regularity.

2.3. **An explicit global continuous metric.** Return to the ruled surface S and the line bundle L in (1.2). Choose a positive smooth metric h_K on K. Using (2.3), extend the dual metric h_K^{-1} on $\mathcal{O}_S(Y)|_Y$ to a smooth metric h_Y on $\mathcal{O}_S(Y)$. Set

$$h_0 = h_Y \otimes \pi^* h_K, \qquad \theta_0 = \Theta_{h_0}(L), \qquad q = \|s_Y\|^2_{h_Y}. \tag{2.15}$$

The metric $h_0|_Y$ is the flat metric under the canonical trivialization of $L|_Y$, and

$$\theta_0|_{TY} = 0, \qquad \Theta_{h_Y}(\mathcal{O}(Y))|_{TY} = -\Theta_{h_K}(K) < 0. \tag{2.16}$$

Fix $0 < a \le \frac{1}{2}$ and choose $\Lambda > 0$ as in Lemma 2.2. Since q vanishes exactly on Y and S is compact, there exists $\varepsilon > 0$ such that the weights $\varphi_0 + \Lambda q^a$ are psh on $\{q < \varepsilon\}$. Write $Q = \max_S q$ and choose $0 < \varepsilon < Q$. Choose a constant

$$B > \max_{\varepsilon \le t \le Q} (\Lambda t^a - \log t). \tag{2.17}$$

Define, on $[0, Q]$,

$$\chi(0) = 0, \qquad \chi(t) = \max\{\Lambda t^a, \log t + B\} \quad (t > 0), \tag{2.18}$$

and put

$$h = h_0 \exp(-\chi(q)). \tag{2.19}$$

Proposition 2.4. *The metric (2.19) is a positive continuous Hermitian metric on L with psh local weights. It is locally $C^{0,2a}$, and is locally Lipschitz for $a = \frac{1}{2}$.*

Proof. The logarithmic branch in (2.18) tends to $-\infty$ as $t \to 0$, so $\chi(t) = \Lambda t^a$ for all sufficiently small t. Hence χ is continuous at zero and bounded on $[0, Q]$.

On $S \setminus Y$, Poincaré–Lelong and (2.15) give

$$\theta_0 + dd^c \log q = \pi^* \Theta_{h_K}(K) \ge 0. \tag{2.20}$$

As a current on S, the same expression equals $\pi^*\Theta_{h_K}(K) + [Y]$. Thus $\varphi_0 + \log q + B$ is a singular psh weight of L.

On $\{0 < q < \varepsilon\}$, the weights of h are

$$\varphi_0 + \chi(q) = \max\{\varphi_0 + \Lambda q^a, \varphi_0 + \log q + B\},$$

a maximum of two psh functions. They are therefore psh. On $\{q \ge \varepsilon\}$, (2.17) makes the logarithmic branch strictly dominant. This domination also holds on a collar of $\{q = \varepsilon\}$, so no boundary gluing issue arises. The weights are psh there by (2.20). Near Y, the cusp branch is dominant, and Lemma 2.2 proves positivity across Y.

All the weights are finite and continuous. Near Y their only nonsmooth term is $\Lambda q^a = \Lambda |w|^{2a} e^{-ab}$, which is $C^{0,2a}$ for $0 < 2a \le 1$. Away from Y, both

branches are smooth, and their maximum is locally Lipschitz, hence locally $C^{0,2a}$. This proves the asserted regularity. □

Remark 2.5 (Bounded potentials and minimal singularities). The current

$$T = \theta_0 + dd^c\chi(q) \in c_1(L)$$

has locally Lipschitz potentials when $a = \frac{1}{2}$. It also has minimal singularities: if $T' = \theta_0 + dd^c v \geq 0$ is another positive current in $c_1(L)$, its quasi-psh potential v is bounded above on compact S, while $\chi(q)$ is bounded below. Hence

$$v \leq \chi(q) + C_v$$

for a constant C_v. This is precisely the comparison expressing that T is no more singular than T'. Here global potentials exist by the $\partial\bar{\partial}$-lemma on the projective surface.

For this modified Grauert example, Proposition 2.4 therefore gives an affirmative answer to the bounded-potential question in [3, Section 4] for the modified example. That question also mentions Grauert's original example; the present argument does not identify its line bundle with L and does not settle that other case. The metric is constructed here, not taken from that preprint. This explicit answer for L is not a statement about arbitrary nef and big line bundles, nor a claim that every minimal-singularity representative has continuous potentials.

3. Line-bundle exhaustions and the Kähler counterexample

3.1. Dual norms and compact-group averaging.

Lemma 3.1. *Let Z be a compact complex manifold and $A \to Z$ a holomorphic line bundle with a positive continuous Hermitian metric h having psh local weights. Then the squared dual norm*

$$\rho(v) = \|v\|^2_{h^{-1}} \qquad (v \in \mathrm{Tot}(A^{-1}))$$

is a continuous psh exhaustion. If the weights are locally $C^{0,\gamma}$ for $0 < \gamma \leq 1$, so is ρ.

Proof. For a local frame e_i of A, write a dual vector as $v = \lambda_i e_i^*$, and write $h(e_i, e_i) = e^{-\varphi_i}$. Then

$$\rho = |\lambda_i|^2 e^{\varphi_i}. \tag{3.1}$$

Off the zero section, $\log\rho = \log|\lambda_i|^2 + \varphi_i$ is psh. Composition with the increasing convex function exp makes ρ psh there. Formula (3.1) extends continuously by zero to the zero section and is locally bounded, so removability gives psh on the whole total space.

Fix a smooth Hermitian metric h_{ref} on A^{-1}. By compactness of Z and positivity of h, constants $c, C > 0$ satisfy

$$c\|v\|^2_{h_{\mathrm{ref}}} \leq \rho(v) \leq C\|v\|^2_{h_{\mathrm{ref}}}.$$

The closed disk bundle of any fixed radius over Z is compact. Every sublevel of ρ is a closed subset of such a disk bundle and is therefore compact. The Hölder assertion follows locally from (3.1). □

Lemma 3.2 (Averaging preserves exhaustions)**.** *Let a compact group G act continuously on a locally compact Hausdorff space X, and let F be a continuous exhaustion. Its normalized Haar average*

$$\overline{F}(x) = \int_G F(gx)\, dg$$

is also an exhaustion. If G is a compact Lie group, X is a complex manifold, the action is smooth and holomorphic, and F is C^2 psh, then $\overline{F}$ is C^2 psh. The same assertion holds in the smooth category.

Proof. For $K_c = \{F \le c\}$, the orbit saturation $G\cdot K_c$ is compact. If $x \notin G\cdot K_c$, then $F(gx) > c$ for every $g \in G$, and continuity on the compact orbit implies $\overline{F}(x) > c$. Hence

$$\{\overline{F} \le c\} \subset G \cdot K_c.$$

The sublevel on the left is closed and therefore compact. Differentiation under the integral preserves C^2 regularity, and averaging the nonnegative Levi forms of the holomorphic pullbacks preserves psh. □

3.2. **The radius criterion.**

Proposition 3.3 (The radius criterion)**.** *Let Z be a compact complex manifold and $A \to Z$ a holomorphic line bundle. If $\mathrm{Tot}(A^{-1})$ admits a C^2 psh exhaustion, then A admits a C^2 Hermitian metric with semipositive curvature. If the exhaustion is smooth, the metric is smooth.*

Proof. Average the exhaustion over the holomorphic circle action $v \mapsto e^{i\alpha}v$ on the fibers, using Lemma 3.2. Denote the resulting invariant exhaustion by F. For a local frame e_i of A and $v = \lambda_i e_i^*$, circle invariance gives

$$F(z,\lambda_i) = U_i(z,t), \qquad t = \log|\lambda_i|^2, \quad \lambda_i \neq 0. \tag{3.2}$$

The function

$$V_i(z,\zeta) = U_i(z, \operatorname{Re}\zeta) = F(z, e^{\zeta/2}) \tag{3.3}$$

is a holomorphic pullback of F and is psh. In particular $U_{i,tt} \ge 0$.

For fixed z, continuity at the zero section gives the finite limit

$$\lim_{t\to-\infty} U_i(z,t) = F(z,0).$$

Exhaustiveness gives $U_i(z,t) \to +\infty$ as $t \to +\infty$. A differentiable convex function with finite left limit has nonnegative derivative. Moreover, if its derivative vanishes at t_0, convexity forces it to be constant on $(-\infty, t_0]$. Consequently

$$U_{i,t}(z,t) > 0 \quad \text{whenever } U_i(z,t) > F(z,0). \tag{3.4}$$

Fix $c > \max_Z F|_Z$. By convexity, the two limits, and (3.4), there is a unique solution

$$U_i(z, \sigma_i(z)) = c, \qquad U_{i,t}(z, \sigma_i(z)) > 0. \tag{3.5}$$

The implicit function theorem makes σ_i of class C^2, or smooth when F is smooth.

We verify the sign of its Levi form explicitly. Differentiating (3.5) twice in base coordinates gives

$$\begin{aligned} 0 = {} & U_{i,z_j\bar{z}_k} + U_{i,z_jt}\sigma_{i,\bar{z}_k} + U_{i,t\bar{z}_k}\sigma_{i,z_j} \\ & + U_{i,tt}\sigma_{i,z_j}\sigma_{i,\bar{z}_k} + U_{i,t}\sigma_{i,z_j\bar{z}_k}, \end{aligned} \tag{3.6}$$

with all derivatives of U_i evaluated at $(z, \sigma_i(z))$. For V_i in (3.3), its $\zeta\bar{\zeta}$ entry is $U_{i,tt}/4$, and its mixed entries have a factor $1/2$. Thus (3.6) is precisely the identity

$$\mathcal{L}_{V_i}(\xi, 2\partial\sigma_i(\xi)) = -U_{i,t}\mathcal{L}_{\sigma_i}(\xi). \tag{3.7}$$

The left side is nonnegative and $U_{i,t} > 0$, so $\psi_i = -\sigma_i$ is psh.

If $e_j = g_{ij}e_i$, the dual coordinates satisfy $\lambda_j = g_{ij}\lambda_i$. The level in (3.5) consequently transforms by

$$\sigma_j = \sigma_i + \log|g_{ij}|^2,$$

so

$$\psi_j = \psi_i - \log|g_{ij}|^2.$$

These are exactly the metric transition laws (1.3). Hence $\|e_i\|^2 = e^{-\psi_i}$ defines the desired semipositive Hermitian metric on A. □

Corollary 3.4. *For a holomorphic line bundle A over a compact complex manifold, $\mathrm{Tot}(A^{-1})$ admits a C^2 psh exhaustion if and only if A admits a C^2 semipositive Hermitian metric. The same equivalence holds with C^2 replaced by C^∞.*

Proof. The forward implication is Proposition 3.3. For the reverse implication, the squared dual norm in Lemma 3.1 has the regularity of the given metric, by its local expression (3.1). □

3.3. **Proof of the Kähler counterexample.**

Proof of Theorem 1.1. The construction in Subsection 2.1 gives a smooth projective surface S and an algebraic line bundle L. The total space M is the open subset

$$M = \mathbf{P}_{\text{lines}}(\mathcal{O}_S \oplus L^{-1}) \setminus \mathbf{P}_{\text{lines}}(L^{-1}). \tag{3.8}$$

The identification sends $v \in L_z^{-1}$ to the line $[(1, v)]$. The projective bundle in (3.8) is a smooth projective threefold. Restricting any of its Kähler forms proves that M is Kähler.

Let h be the continuous semipositive metric from Proposition 2.4. If $p : M \to S$ is the bundle projection, the exhaustion is explicitly

$$\rho(v) = \|v\|_{h^{-1}}^2 = \|v\|_{h_0^{-1}}^2 \exp\bigl(\chi(q(p(v)))\bigr). \tag{3.9}$$

Lemma 3.1 shows that ρ is continuous, psh, and exhaustive. Taking $a = \frac{1}{2}$ and the coefficient Λ supplied by Lemma 2.2 makes h, its weights, and ρ locally Lipschitz. In particular the same example has local $C^{0,\gamma}$ regularity for all $0 < \gamma \leq 1$.

If a C^2 psh exhaustion on M existed, Proposition 3.3 would produce a C^2 semipositive Hermitian metric on L. Proposition 2.1 excludes such a metric. Therefore no C^2 psh exhaustion exists, completing the proof. □

Remark 3.5 (Interpretation). The example separates existence of a continuous exhaustion from existence of any smooth exhaustion. The conclusion does not depend on whether the particular function (3.9) can be smoothed: every hypothetical C^2 exhaustion leads to the forbidden metric. The argument does not determine whether a C^1 psh exhaustion exists. Corollary 5.7 shows that dimension three is minimal within smooth quasi-projective manifolds; minimality among arbitrary noncompact Kähler manifolds is not established. Thus a definition of weak pseudoconvexity requiring a smooth psh exhaustion is strictly stronger than the corresponding definition requiring only a continuous psh exhaustion, even for smooth quasi-projective Kähler threefolds.

4. Compact modifications and non-Kähler counterexamples

The Kähler counterexample can be changed over a single point to obtain a non-Kähler counterexample of the same dimension. We first isolate the exhaustion argument, then insert Hironaka's opposite-order blowup configuration in an exceptional plane. Compact products give a simpler alternative when an increase in dimension is allowed.

4.1. **Exhaustions under compactly supported modifications.**

Proposition 4.1. *Let $f : X' \to X$ be a proper surjective holomorphic map between nonempty complex manifolds. Suppose that, for a compact set $K \subset X$, it restricts to a biholomorphism*

$$X' \setminus f^{-1}(K) \longrightarrow X \setminus K.$$

For every $r \in \{0, 2, \infty\}$, X admits a C^r psh exhaustion if and only if X' does.

Proof. The case $K = \varnothing$ is immediate. Holomorphic pullback preserves plurisubharmonicity and the stated regularity, and properness makes the sublevels of a pulled-back exhaustion compact. Conversely, suppose F is a C^r psh exhaustion on X'. Choose a relatively compact open neighborhood U of K and

$$c > \max_{f^{-1}(\overline{U})} F.$$

Choose a smooth convex nondecreasing function $\chi : \mathbb{R} \to [0, \infty)$ that vanishes on $(-\infty, c]$ and tends to infinity. For example, integrate a smooth nondecreasing function which is zero up to c and one from $c+1$ onward. On $X \setminus K$ define

$$G = \chi \circ F \circ (f|_{X' \setminus f^{-1}(K)})^{-1}.$$

This function is C^r and psh. It vanishes on $U \setminus K$, so extension by zero on K gives a C^r psh function on all of X. For each nonempty sublevel $\{G \leq a\}$ choose b such that $\chi(t) \leq a$ implies $t \leq b$. Then

$$\{G \leq a\} \subset K \cup f(\{F \leq b\}).$$

The left side is closed and the right side is compact. Thus G is an exhaustion; no smooth descent of the original function F is needed. □

4.2. **A non-Kähler modification over one point.**

Proposition 4.2. *Let X be a connected smooth complex threefold and $x \in X$. There is a connected smooth non-Kähler threefold N and a proper holomorphic modification $\mu : N \to X$ inducing a biholomorphism*

$$N \setminus \mu^{-1}(x) \simeq X \setminus \{x\}.$$

Proof. Blow up x, obtaining

$$b : W = \mathrm{Bl}_x X \longrightarrow X, \qquad E = b^{-1}(x) \simeq \mathbf{P}^2.$$

In homogeneous coordinates $[u : v : w]$ on E, take

$$A = \{w = 0\}, \qquad B = \{uv = w^2\}.$$

The line A and smooth conic B meet transversely at exactly $p = [1 : 0 : 0]$ and $q = [0 : 1 : 0]$. Near either point they are the two coordinate axes in a smooth surface E inside W.

Over $W \setminus \{q\}$, first blow up $A \setminus \{q\}$ and then the strict transform of $B \setminus \{q\}$. Over $W \setminus \{p\}$, first blow up $B \setminus \{p\}$ and then the strict transform of $A \setminus \{p\}$. All these centers are closed smooth curves in the respective open manifolds. On $W \setminus \{p, q\}$ the centers are disjoint, so the blowups commute canonically over the base. Glue the two resulting manifolds through this isomorphism and write

$$\nu : N \longrightarrow W, \qquad \mu = b \circ \nu.$$

The glued space is Hausdorff: points with distinct images can be separated in W, and points with the same image belong to one of the Hausdorff open manifolds lying over the two base opens. It is smooth of dimension three because every blowup center is smooth. Properness is local on the target, so ν and μ are proper. Each blowup chart is connected and the charts overlap, hence N is connected. All centers lie in E, so μ is an isomorphism away from x. This is Hironaka's opposite-order construction [16, Section 6]; see also [17, Tag 08J1].

For completeness, the obstruction to a Kähler form is entirely contained in compact exceptional surfaces. Let $D_A, D_B \subset N$ be the exceptional divisors

dominating A, B, and let $\alpha, \beta \in H_2(N, \mathbb{R})$ be the classes of their general $\mathbf{P}^1$ fibers. The surfaces D_A, D_B are smooth and compact: locally each is a ruled surface or its blowup at a point, and they map properly to the compact curves A, B. The fiber divisors of each map are linearly equivalent, since $A \simeq B \simeq \mathbf{P}^1$, and hence are homologous on the corresponding compact surface.

Over p, the strict transform of B meets the first exceptional divisor transversely at one point. The second blowup therefore restricts on that divisor to a surface blowup at that point. Its fiber over p is the reduced union of the strict transform $R_p \simeq \mathbf{P}^1$ of the first exceptional fiber and the new exceptional fiber $T_p \simeq \mathbf{P}^1$. Both occur with multiplicity one. Indeed, take local coordinates (x, y, z) with $A = \{y = z = 0\}$ and $B = \{x = z = 0\}$. In the first blowup chart $z = yt$, the first exceptional divisor is $y = 0$ and the second center is $x = t = 0$. Its restriction to $y = 0$ is the blowup of the smooth surface with coordinates (x, t) at the origin. The pullback of the fiber equation $x = 0$ is its strict transform plus the exceptional curve, each once. The curve T_p is also the fiber of $D_B \to B$ over p. Thus

$$\alpha = [R_p] + [T_p], \qquad \beta = [T_p].$$

Over q the order is reversed. With analogous notation,

$$\beta = [R_q] + [T_q], \qquad \alpha = [T_q].$$

These relations hold in ordinary homology: they arise from fiber divisors on compact D_A and D_B, regardless of compactness of N. Adding them gives

$$[R_p] + [R_q] = 0 \quad \text{in } H_2(N, \mathbb{R}). \tag{4.1}$$

If ω were a Kähler form on N, closedness and (4.1) would imply

$$0 = \int_{R_p} \omega + \int_{R_q} \omega > 0,$$

a contradiction. □

Proof of Theorem 1.2. Apply Proposition 4.2 to the threefold M of Theorem 1.1 at the chosen point x. If ρ is its continuous psh exhaustion, then

$$\rho_N = \rho \circ \mu, \qquad \{\rho_N \le a\} = \mu^{-1}(\{\rho \le a\})$$

is a continuous psh exhaustion of N. A holomorphic map is locally Lipschitz in smooth coordinate charts, so local Lipschitz regularity of the exhaustion is preserved. A C^2 psh exhaustion on N would give one on M by Proposition 4.1 with $K = \{x\}$, contrary to Theorem 1.1. □

Remark 4.3 (The sense in which the change is local). The point x can be chosen outside the zero section of $M \to S$. The construction then preserves a neighborhood of that section and, by properness, a neighborhood of infinity. More precisely, for every neighborhood U of x, μ is an isomorphism over $M \setminus U$. This is a modification supported over one point; it makes no claim about a small deformation of the complex structure on a fixed smooth manifold.

4.3. **Compact products and an explicit fivefold.**

Proposition 4.4 (Compact product principle)**.** *Let X be a nonempty complex manifold and T a nonempty compact connected complex manifold. For each $r \in \{0, 2, \infty\}$, X admits a C^r psh exhaustion if and only if $X \times T$ does. If T is non-Kähler, then $X \times T$ is non-Kähler.*

Proof. Pullback by the projection preserves regularity and plurisubharmonicity, with compact sublevels

$$\{f \circ \mathrm{pr}_X \leq a\} = \{f \leq a\} \times T. \tag{4.2}$$

Conversely, restriction to the closed holomorphic slice $X \times \{t_0\}$ preserves regularity, plurisubharmonicity, and compact sublevels. A Kähler form on the product would restrict to one on $\{x_0\} \times T$, proving the final assertion. In fact, the maximum principle makes every continuous psh function on the product constant on each compact connected fiber $\{x\} \times T$. □

Corollary 4.5 (A non-Kähler fivefold)**.** *Let M be the threefold of Theorem 1.1 and let*

$$H = (\mathbb{C}^2 \setminus \{0\})/\langle z \mapsto 2z \rangle. \tag{4.3}$$

Then $M \times H$ is a smooth non-Kähler fivefold admitting a continuous psh exhaustion but no C^2 psh exhaustion. For every $0 < \gamma < 1$ its exhaustion can be chosen locally $C^{0,\gamma}$.

Proof. The dilation action is free and properly discontinuous: if a compact set lies in $r \leq \|z\| \leq R$, only finitely many integers n satisfy $r/R \leq 2^n \leq R/r$. Polar coordinates identify its quotient smoothly with

$$H \simeq S^3 \times (\mathbb{R}/(\log 2)\mathbb{Z}) \simeq S^3 \times S^1. \tag{4.4}$$

Thus H is a compact connected complex surface with $H^2_{\mathrm{dR}}(H) = 0$. Any Kähler form ω would be exact, and Stokes' theorem would give $\int_H \omega^2 = 0$, contradicting positivity. Apply Proposition 4.4 and Theorem 1.1. Explicitly,

$$\rho_{M \times H}(v, [z]) = \rho(v) \tag{4.5}$$

is the required exhaustion, with the same local Hölder regularity. □

5. Surface regularization and boundary existence criteria

We prove that a continuous psh exhaustion on a divisor complement in a compact Kähler surface implies a smooth one, and hence obtain regularization for every smooth complex quasi-projective surface. The averaging and current argument below treats every reducible or disconnected boundary and requires no minimal-model reduction. We distinguish this conditional regularization theorem from the separate numerical and Ueda-theoretic criteria that construct an exhaustion without assuming a continuous one.

Unless otherwise specified, a divisor compactification means $X = S \setminus \operatorname{supp} D$, where S is a smooth compact connected complex surface and $D \neq 0$

is effective. The Enriques–Kodaira classification concerns S, not X. Table 1 summarizes the consequences for such compactifiers. No divisorial compactification is assumed to exist for an arbitrary noncompact surface.

5.1. **Two basic affirmative cases.**

Proposition 5.1 (A known holomorphic-convexity consequence)**.** *Let X be a connected noncompact complex manifold of dimension two with a continuous psh exhaustion. If $\mathcal{O}(X) \neq \mathbb{C}$, then X has a smooth psh exhaustion.*

Proof. The holomorphic-convexity theorem of Vîjîitu [4, Theorem 1.1] makes X holomorphically convex. Its Remmert reduction $r : X \to Y$ is proper with Stein target. A smooth strongly psh exhaustion of the complex space Y pulls back to a smooth psh exhaustion of X. Here smoothness on Y means local restriction of smooth ambient functions, so the pullback is smooth even over singular points of Y. □

This is a consequence of the cited theorem, not a new holomorphic-convexity result. This proposition alone leaves the case $\mathcal{O}(X) = \mathbb{C}$ untreated; Theorem 5.4 below also covers that case under its compactification hypothesis. Likewise, the regularization theorem of Mongodi–Slodkowski for domains in a surface already carrying a real-analytic psh exhaustion [5, Corollaries 4.4–4.5] requires that ambient hypothesis; one cannot assume it for X itself.

Proposition 5.2 (Line-bundle total spaces over a curve)**.** *Let A be a holomorphic line bundle on a compact connected Riemann surface B. The following are equivalent:*

$$\mathrm{Tot}(A^{-1}) \text{ has a continuous psh exhaustion}$$
$$\iff \quad \deg A \geq 0 \quad \iff$$
$$\mathrm{Tot}(A^{-1}) \text{ has a smooth psh exhaustion.}$$

Proof. Choose a smooth reference metric with curvature θ_0 and an area form ω with $\int_B \omega = 1$. For $d = \deg A \geq 0$, solve $dd^c f = d\omega - \theta_0$. The resulting metric has curvature $d\omega$, and its squared dual norm is a smooth psh exhaustion by Lemma 3.1. Conversely, the compactification

$$\mathrm{Tot}(A^{-1}) = \mathbf{P}_{\mathrm{lines}}(\mathcal{O}_B \oplus A^{-1}) \setminus \mathbf{P}_{\mathrm{lines}}(A^{-1})$$

has infinity section with normal bundle A. If $\deg A < 0$, that section has negative self-intersection, contradicting the boundary obstruction in Lemma 5.3(i) below. □

Thus the line-bundle construction of Theorem 1.1 cannot be lowered to dimension two by replacing its compact base by a curve. The full quasi-projective surface case is covered by Corollary 5.7.

5.2. **Boundary tools and permitted modifications.** For a divisor complement, a continuous exhaustion tends uniformly to $+\infty$ at the boundary: each sublevel is a compact subset of X, hence its complement in S is a neighborhood of $\operatorname{supp} D$. We call a psh function on a punctured boundary neighborhood which diverges in this way a *divergent collar*.

Blowing up a boundary point preserves X only when the new boundary is the *full reduced inverse image*, including the exceptional curve. This permits resolution of the boundary without changing X. Blowups at interior points give proper modifications with compact exceptional locus, and Proposition 4.1 preserves both exhaustion properties. By contrast, contracting a curve which meets but is not contained in the boundary need not induce a proper map of the complements. We make no unrestricted reduction of projective pairs to minimal compactifiers.

Lemma 5.3 (Boundary obstruction and collar construction)**.** *Let S be smooth compact and let $D \neq 0$ be effective.*

(i) *If a connected component of* $\operatorname{supp} D$ *has negative definite intersection matrix, then $S \setminus \operatorname{supp} D$ has no continuous psh exhaustion.*

(ii) *If every connected boundary component admits a smooth divergent psh collar, then $S \setminus \operatorname{supp} D$ has a smooth psh exhaustion.*

(iii) *Let $E = \bigcup E_i$ be a connected reduced simple normal crossing* (SNC) *divisor and let λ be the largest eigenvalue of $I = (E_i \cdot E_j)$. If $\lambda > 0$, E admits a smooth strictly psh divergent collar. If $\lambda = 0$, the matrix is negative semidefinite with one-dimensional kernel, and there are positive integers a_i with*

$$A = \sum_i a_i E_i, \qquad A \cdot E_i = 0, \qquad A^2 = 0. \tag{5.1}$$

When S is projective, this divisor A is nef.

Proof. For (i), Grauert's contraction theorem [6, §4, subsection 8(e), pp. 366–367] contracts the indicated component to a point p of a normal surface, without changing a punctured neighborhood. A continuous psh function on such a punctured neighborhood is bounded above near p. Indeed, a local finite projection to a bidisc, with p the only point over 0, sends it to the fiberwise maximum v_*, a continuous psh function on the punctured bidisc. This maximum is psh off the branch locus; continuity and local boundedness extend this across the branch locus away from 0. On

$$\{|z_1| \leq r,\ |z_2| = r\}$$

it has a finite maximum. The maximum principle on each vertical disc with $z_1 \neq 0$, followed by continuity at $z_1 = 0$, gives the same bound on the punctured smaller bidisc. This contradicts divergence of an exhaustion at the contracted boundary.

For (ii), choose disjoint smaller boundary neighborhoods W_E compactly contained in the collar neighborhoods and choose T_E larger than the collar function on ∂W_E. A smooth convex nondecreasing function χ_E, zero on

$(-\infty, T_E]$ and tending to infinity, makes $\chi_E(\Phi_E)$ vanish near ∂W_E. Extend by zero outside W_E and sum over the finitely many boundary components. The result is smooth and psh, diverges at every boundary point, and is proper because S is compact.

For (iii), the off-diagonal entries of I are nonnegative and its graph is connected. Perron–Frobenius applied after a scalar shift makes λ simple with a strictly positive eigenvector. If $\lambda > 0$, rational approximation and clearing denominators give $A = \sum a_i E_i$ with $a_i > 0$ and $A \cdot E_i > 0$ for every i. We spell out the metric construction needed here. The restriction $\mathcal{O}_S(A)|_E$ is ample. Embed E by a power of this line bundle and use its Fubini–Study metric. Lift the embedding sections locally to S and take the appropriate metric root. Their ambient extensions have positive tangential curvature along the smooth part of E; at a node they have positive curvature in all directions, since the Zariski tangent space of E there is the ambient tangent space and the embedding is injective on tangent spaces. Patch these extensions using a partition of unity relative to a reference metric, keeping the chosen metrics near the nodes. The restriction to E, and hence the tangential curvature on its smooth part, is unchanged.

Let $q = \|s_E\|^2$ for any smooth metric on $\mathcal{O}_S(E)$. On the smooth part of E, $dd^c q$ is positive in the normal direction and zero in the tangent direction; at a node it is zero. Adding Cq to the patched weights, for sufficiently large C, makes the curvature strictly positive along all of E, by the Schur complement and compactness, and hence in a neighborhood. For this metric h, $-\log \|s_A\|_h^2$ is the required collar.

If $\lambda = 0$, simplicity gives a one-dimensional positive kernel, which has a rational generator because I is integral. Clearing denominators proves (5.1). Intersections with curves outside E are nonnegative by effectiveness; in the projective case this proves nefness. □

In particular, under the continuous-exhaustion hypothesis a connected SNC boundary is either already handled by a positive collar, or supports the weighted square-zero divisor (5.1). A negative definite boundary is an *exclusion*, not a continuous-versus-smooth counterexample.

5.3. **Regularization for divisor complements in compact Kähler surfaces.** The continuous-to-smooth implication for divisor complements follows from positive dd^c-closed currents on compact Kähler surfaces. The argument applies to reducible, disconnected, and singular divisors and does not require a minimal compactifier. We use the normalization of Section 1.5. A modification of a Stein space means a proper holomorphic modification onto a Stein space.

Theorem 5.4 (Regularization for compact Kähler divisor complements)**.** *Let S be a smooth compact connected Kähler surface and let $D \neq 0$ be a reduced effective divisor. If $X = S \setminus D$ admits a continuous plurisubharmonic exhaustion, then it admits a smooth plurisubharmonic exhaustion.*

More precisely, if X is not a modification of a Stein space, there are positive integers a_i, one for every irreducible component D_i of D, such that

$$A = \sum_i a_i D_i, \qquad A \cdot D_i = 0 \text{ for every } i, \qquad A^2 = 0,$$

and $\mathcal{O}_S(A)$ admits a smooth Hermitian metric with zero curvature on a neighborhood of D. In particular, a punctured neighborhood of the entire boundary admits a smooth pluriharmonic function tending to $+\infty$ along D.

We first prove the averaging statement that supplies the current used in the proof. Its conclusion concerns Lelong numbers at every point, including points at which a weak limit could acquire new concentration.

Lemma 5.5 (Currents obtained by averaging exhaustion levels)**.** *Let (S, ω) be a compact Kähler surface, and let $X \subset S$ be a connected noncompact open subset with a continuous plurisubharmonic exhaustion ρ. If X is not a modification of a Stein space, there is a positive $(1,1)$-current T on S such that*

(5.2)

$$dd^c T = 0, \qquad \int_S T \wedge \omega = 1, \qquad \operatorname{supp} T \Subset X, \qquad \nu(T, x) = 0 \quad (x \in S).$$

Proof. Let Σ_X^0 be the continuous minimal kernel: the intersection, over all continuous plurisubharmonic exhaustions ψ of X, of the sets where ψ is not strongly plurisubharmonic. Here strong plurisubharmonicity means local stability under sufficiently small C^2 perturbations. By [5, Lemma 2.1], if $\rho(\Sigma_X^0)$ contains no upper half-line, then X is a modification of a Stein space. Thus, under the present hypothesis, there is $t_0 \in \mathbb{R}$ such that

$$(t_0, +\infty) \subset \rho(\Sigma_X^0). \tag{5.3}$$

This invocation uses the continuous version of that lemma, which is stated for arbitrary complex manifolds.

Fix $t > t_0$ and put $K_t = \{\rho = t\}$. We claim that no smooth strictly plurisubharmonic function exists on a neighborhood of the whole compact set K_t. Suppose, to the contrary, that v is such a function on a neighborhood W. Properness of ρ allows us to choose $\delta > 0$ with

$$B = \{t - \delta \le \rho \le t + \delta\} \Subset W.$$

After a positive affine change of v, assume $0 < v < 1$ on B. Choose $0 < \eta < 1$, $0 < \varepsilon < \eta\delta/2$, and set $g(s) = s + \eta|s - t|$. The function g is convex, increasing, and tends to $+\infty$ as $s \to +\infty$. Define on X

$$\psi(x) = \begin{cases} \max\{g(\rho(x)),\, \rho(x) + \varepsilon v(x)\}, & |\rho(x) - t| < \delta, \\ g(\rho(x)), & |\rho(x) - t| \ge \delta. \end{cases}$$

On the part of the band where $|\rho - t| > \delta/2$, the first branch strictly dominates, so the definitions glue. On K_t the second branch strictly dominates, and hence it also dominates on a neighborhood of K_t. Consequently ψ is a

continuous plurisubharmonic exhaustion strongly plurisubharmonic near K_t. This contradicts $K_t \cap \Sigma_X^0 \neq \varnothing$, which follows from (5.3).

We now apply Hahn–Banach separation on K_t. Let $\mathcal{E}_t$ be the real Banach space of continuous sections over K_t of the bundle of real $(1,1)$-forms on S. Its open convex cone $\mathcal{P}_t$ consists of forms that are positive definite at every point of K_t. The linear subspace

$$\mathcal{L}_t = \{dd^c f|_{K_t} : f \in C^\infty(S, \mathbb{R})\}$$

does not meet $\mathcal{P}_t$, by the preceding claim. Conversely, a smooth function defined near K_t can be extended to a smooth function on S agreeing with it near K_t, by a cutoff; thus there is no distinction between the local and global formulations here. Separating the linear subspace from the open cone gives a nonzero continuous real linear functional Λ_t vanishing on $\mathcal{L}_t$ and nonnegative on $\mathcal{P}_t$, and hence on its closure. The subspace need not be closed for this separation. Restriction of test forms to K_t turns Λ_t into an order-zero positive $(1,1)$-current supported on K_t. Its annihilation of $dd^c f$ for all smooth f says that it is dd^c-closed. Its pairing with ω is strictly positive: otherwise positivity and the inequalities $-c\omega \leq \gamma \leq c\omega$ on K_t would make it vanish on every $\gamma \in \mathcal{E}_t$. We may therefore normalize it to obtain

$$T_t \geq 0, \qquad dd^c T_t = 0, \qquad \operatorname{supp} T_t \subset K_t, \qquad \int_S T_t \wedge \omega = 1. \tag{5.4}$$

Fix $t_0 < a < b$. For $n \geq 1$ choose the equally spaced levels

$$t_{n,j} = a + \frac{j - \frac{1}{2}}{n}(b-a), \qquad 1 \leq j \leq n, \qquad T_n = \frac{1}{n}\sum_{j=1}^{n} T_{t_{n,j}}.$$

Only finitely many arbitrary choices are needed for each n; no measurable selection is involved. Every T_n has mass one and is supported in the fixed compact shell $K = \rho^{-1}([a,b]) \Subset X$. Weak compactness of positive currents of bounded mass on S yields a subsequence, still denoted by T_n, converging to a positive current T. Testing against ω and against $dd^c f$ shows that T has mass one and is dd^c-closed. Testing on $S \setminus K$ shows $\operatorname{supp} T \subset K$.

It remains to rule out Lelong concentration. Fix $x \in X$ and choose holomorphic coordinates on a neighborhood $U \Subset X$ containing $\overline{B(x,2R)}$. Write $\beta = dd^c|z|^2$. The Lelong–Jensen monotonicity for positive dd^c-closed currents, recalled in [18, Appendix B, equations (B.1)–(B.2)], implies that for every current in (5.4) and $0 < r < R$,

$$\frac{1}{r^2}\int_{B(x,r)} T_t \wedge \beta \leq \frac{1}{R^2}\int_{B(x,R)} T_t \wedge \beta \leq C_R. \tag{5.5}$$

Indeed, $\beta \leq c\omega$ on $\overline{B(x,R)}$, so one may take $C_R = c/R^2$ using the mass normalization. Thus C_R is independent of t, n, and j. Monotonicity applies to open balls as well; alternatively it follows for them by increasing limits from smaller radii. The positive factor between the dd^c convention in [18] and ours changes neither this monotonicity nor vanishing of Lelong numbers.

If $T_{t_{n,j}}$ contributes mass to $B(x,r)$, then its supporting level belongs to

$$I_r = \left[\inf_{B(x,r)} \rho,\ \sup_{B(x,r)} \rho \right].$$

The number of grid points in I_r is at most $n|I_r|/(b-a)+2$. Hence (5.5) gives

$$\frac{1}{r^2} \int_{B(x,r)} T_n \wedge \beta \le C_R \left(\frac{\operatorname{osc}_{B(x,r)} \rho}{b-a} + \frac{2}{n} \right). \tag{5.6}$$

For completeness, choose $\chi \in C_c^\infty(U)$ nonnegative and equal to one on $B(x,R)$. Extend $\chi\beta$ by zero to S. The positive measures $\mu_n = T_n \wedge \chi\beta$ converge weakly to $\mu = T \wedge \chi\beta$: current convergence gives convergence on smooth test functions, and the uniform mass bound extends this to continuous test functions. The open-set inequality for weak convergence of measures gives, for the open ball $B(x,r)$,

$$\int_{B(x,r)} T \wedge \beta = \mu(B(x,r)) \le \liminf_{n\to\infty} \mu_n(B(x,r)).$$

Passing to the limit in (5.6) therefore yields

$$\frac{1}{r^2} \int_{B(x,r)} T \wedge \beta \le \frac{C_R}{b-a} \operatorname{osc}_{B(x,r)} \rho. \tag{5.7}$$

Continuity of ρ makes the right side tend to zero as $r \downarrow 0$. Thus $\nu(T,x) = 0$. At every point of $S \setminus X$ the current vanishes on a neighborhood, since its support is contained in $K \Subset X$. This proves (5.2). The same weakly convergent subsequence works for every fixed ball; no diagonal choice depending on x or r is required. □

We record explicitly the consequence of the surface current theorem that will be used. In particular, a de Rham class is not assigned to a nonclosed current without explanation.

Lemma 5.6 (The surface current inequality)**.** *Let T be a positive dd^c-closed $(1,1)$-current on a compact Kähler surface S, with $\nu(T,x) = 0$ for all $x \in S$. There is a well-defined class $\alpha_T \in H^{1,1}(S,\mathbb{R})$ such that, for every smooth closed real $(1,1)$-form η,*

$$\alpha_T \cdot [\eta] = \int_S T \wedge \eta. \tag{5.8}$$

It satisfies

$$\alpha_T^2 \ge 0, \qquad \alpha_T^2 > 0 \text{ if } dT \ne 0. \tag{5.9}$$

If $dT = 0$, then α_T is the usual cohomology class of T.

Proof. The decomposition in [18, §2.1, equation (2.1), and Lemma B.2] writes

$$T = \Omega + \partial R + \bar\partial \overline{R} + i\partial\bar\partial u,$$

where Ω is a smooth real closed $(1,1)$-form, R is a $(0,1)$-current, and u is a real distribution. Define $\alpha_T = [\Omega]$. Integration by parts gives (5.8). Nondegeneracy of the intersection pairing on $H^{1,1}(S,\mathbb{R})$ makes $[\Omega]$ independent of the decomposition and identifies it with $[T]$ when T is closed.

Since the positive-Lelong-number locus is empty, [18, Corollary 2.4 and its proof] gives (5.9). More specifically, its proof applies [18, Theorem 2.2(2)] to the pair (T,T) and uses the vanishing-energy criterion following equation (B.4). The hypothesis there is that each current gives no mass to the positive-Lelong-number locus of the other, which is automatic here. These results concern arbitrary positive dd^c-closed currents on compact Kähler surfaces; they impose no foliation hypothesis. We use only the qualitative inequalities (5.9), which are independent of the positive normalization factor in dd^c. □

Proof of Theorem 5.4. Fix a Kähler form ω on S and a continuous plurisubharmonic exhaustion of X. If X is a modification $\pi : X \to Y$ of a Stein space, a smooth plurisubharmonic exhaustion of Y pulls back to one on X, since π is proper. Smoothness on a complex space is understood in local embeddings; Stein spaces admit smooth strictly plurisubharmonic exhaustions.

We may therefore suppose that X is not a modification of a Stein space. Lemmas 5.5 and 5.6 give a current T, supported in a compact set $K \Subset X$, and a class $\alpha = \alpha_T \in H^{1,1}(S,\mathbb{R})$ with

$$\alpha \cdot [\omega] = 1, \qquad \alpha^2 \geq 0, \qquad \alpha^2 > 0 \text{ if } dT \neq 0. \tag{5.10}$$

In particular $\alpha \neq 0$.

For each irreducible component D_i, choose a smooth Hermitian metric h_i on $\mathcal{O}_S(D_i)$, with curvature θ_i, and denote its canonical section by s_i. Poincaré–Lelong gives

$$dd^c \log \|s_i\|_{h_i}^2 = [D_i] - \theta_i.$$

Choose a smooth cutoff equal to one on a neighborhood of K and supported away from D_i. Multiplying $\log \|s_i\|_{h_i}^2$ by this cutoff and extending by zero gives a globally smooth function f_i with $dd^c f_i = -\theta_i$ near K. Therefore

$$\alpha \cdot [D_i] = \int_S T \wedge \theta_i = -\langle T, dd^c f_i \rangle = 0. \tag{5.11}$$

This calculation uses only dd^c-closedness of T, not d-closedness, and is legitimate by (5.8).

We now treat all connected boundary components. For such a component E, let $I_E = (D_i \cdot D_j)_{D_i, D_j \subset E}$ and denote its largest eigenvalue by λ_E. If $\lambda_E < 0$, the matrix is negative definite, contrary to Lemma 5.3(i). If $\lambda_E > 0$, a real vector with positive quadratic value can be approximated by a rational vector. Clearing denominators gives an integral divisor B supported on E with $B^2 > 0$. (Its coefficients need not be positive for this argument.) Equation (5.11) gives $\alpha \cdot [B] = 0$. The Hodge index theorem for the compact Kähler surface says that the intersection form on $H^{1,1}(S,\mathbb{R})$ has signature

$(1, h^{1,1}-1)$. Consequently $[B]^{\perp}$ is negative definite, so $\alpha^2 < 0$ because $\alpha \neq 0$. This contradicts (5.10). We conclude that

$$\lambda_E = 0 \qquad \text{for every connected component } E. \tag{5.12}$$

The off-diagonal entries of I_E are nonnegative, and their graph is connected. After adding a sufficiently large scalar multiple of the identity, Perron–Frobenius applies to a nonnegative irreducible matrix. Thus the largest eigenvalue of I_E is simple and has an eigenvector all of whose entries are strictly positive. By (5.12), its kernel is exactly this one-dimensional eigenspace. Since I_E is integral, its kernel has a rational generator. Scaling the positive generator therefore gives positive integers a_i and a divisor

$$A_E = \sum_{D_i \subset E} a_i D_i, \qquad A_E \cdot D_i = 0 \quad (D_i \subset E), \qquad A_E^2 = 0.$$

Set $A = \sum_E A_E$. Distinct connected boundary components are disjoint, so

$$\operatorname{supp} A = D, \qquad A \cdot D_i = 0 \text{ for every } i, \qquad A^2 = 0. \tag{5.13}$$

Every coefficient of A is strictly positive. None of these matrix arguments requires the components to be smooth or the divisor to have normal crossings.

The class $[A]$ is nonzero, since A is nonzero and effective and $\int_A \omega > 0$. Equations (5.11) and (5.13) give $\alpha \cdot [A] = 0$. For a nonzero isotropic vector in a real vector space of signature $(1, h^{1,1}-1)$, its orthogonal complement is negative semidefinite and has radical equal to the line it spans. Hodge index therefore gives $\alpha^2 \leq 0$, with equality only when α is a multiple of $[A]$. Combining this with (5.10) proves

$$dT = 0, \qquad [T] = \alpha = c[A], \qquad c = \frac{1}{\int_A \omega} > 0. \tag{5.14}$$

Only now do we use $[T]$ as the de Rham class of a closed current.

The dd^c-lemma for currents on the compact Kähler manifold S applied to (5.14) yields a real distribution u such that

$$dd^c u = T - c[A]. \tag{5.15}$$

Choose a neighborhood U of D disjoint from K. On a small coordinate neighborhood in U, let f_A be a local holomorphic equation of the effective Cartier divisor A. Then

$$dd^c\big(u + c \log |f_A|^2\big) = 0$$

as distributions on the entire coordinate neighborhood, including its intersection with D. A pluriharmonic distribution is smooth: taking a trace gives a distributional Laplace equation, so elliptic regularity applies. Hence on that neighborhood

$$u = -c \log |f_A|^2 + h, \qquad h \text{ smooth and pluriharmonic.} \tag{5.16}$$

These identities show that u is a smooth pluriharmonic function on $U \setminus D$ and tends to $+\infty$ as one approaches any point of D. The divergence is uniform near D by compactness and a finite coordinate covering. At a

normal crossing, for example, the singular part is $-c(a_i \log|z|^2 + a_j \log|w|^2)$. The same conclusion follows from a local equation f_A at an arbitrary singular point.

To see the asserted metric explicitly, take local frames e_i of $\mathcal{O}_S(A)$ with $s_A = f_i e_i$, and put $b_i = r_i/c$, where $r_i = u + c\log|f_i|^2$ is the smooth pluriharmonic remainder in (5.16). If $e_j = g_{ij}e_i$, then $f_i = g_{ij}f_j$ and

$$b_j = b_i - \log|g_{ij}|^2.$$

Thus $\|e_i\|^2 = \exp(-b_i)$ defines a smooth Hermitian metric on $\mathcal{O}_S(A)|_U$. Its curvature is $dd^c b_i = 0$, and $-\log\|s_A\|^2 = u/c$ on $U \setminus D$.

Finally choose $W \Subset U$ containing D, with compact boundary disjoint from D. Choose M larger than the values of u on a neighborhood of ∂W. Let $\chi : \mathbb{R} \to \mathbb{R}$ be smooth, convex, and nondecreasing, identically zero on $(-\infty, M]$, and tending to $+\infty$ at $+\infty$. Define

$$\Phi = \begin{cases} \chi(u), & \text{on } W \setminus D, \\ 0, & \text{on } X \setminus W. \end{cases}$$

The function vanishes near ∂W, so it is globally smooth. It is plurisubharmonic because u is pluriharmonic on the collar and χ is convex. It diverges along all of D, by the full-support condition in (5.13). Compactness of S then makes every sublevel of Φ compact in X. This completes the proof. □

Corollary 5.7 (Smooth quasi-projective surfaces)**.** *Let X be a smooth complex quasi-projective surface. If X admits a continuous plurisubharmonic exhaustion, then it admits a smooth plurisubharmonic exhaustion.*

Proof. For connected compact X the assertion is immediate. Otherwise take a projective closure $Y \subset \mathbb{P}^N$. Embedded resolution, chosen to preserve the regular locus of Y, gives a smooth projective surface containing the given smooth open subset X. Principalize the ideal of its complement by blowups with smooth centers supported outside X. The full inverse-image boundary is then an effective divisor with simple normal crossings, and its complement is still isomorphic to X; see [19, Theorems 1.0.1–1.0.2]. Thus $X = S \setminus D$ for a smooth projective surface S and a nonempty reduced divisor D. Apply Theorem 5.4. The use of the full inverse image includes any originally zero-dimensional boundary. For a disconnected X, apply the argument to its finitely many connected components. □

Remark 5.8 (Quasi-projective surfaces with only constant holomorphic functions). Corollary 5.7 is not subsumed by Proposition 5.1: a noncompact smooth quasi-projective surface can admit a smooth psh exhaustion while satisfying $\mathcal{O}(X) = \mathbb{C}$, where $\mathcal{O}(X)$ denotes all analytic holomorphic functions, not merely algebraic regular functions. Indeed, let B be a smooth projective curve of positive genus, let $A \in \mathrm{Pic}^0(B)$ be non-torsion, and put $X = \mathrm{Tot}(A^{-1})$. A unitary flat metric on A^{-1} gives a smooth psh exhaustion by its squared fiber norm, as in Proposition 5.2.

For any holomorphic function on X, its degree-n fiberwise Taylor coefficient along the zero section is a section of A^n. For $n > 0$ these sections vanish, since a nontrivial degree-zero line bundle has no nonzero holomorphic section. The degree-zero coefficient is constant on B, so $\mathcal{O}(X) = \mathbb{C}$. The projective completion $S = \mathbf{P}_{\text{lines}}(\mathcal{O}_B \oplus A^{-1})$ has a single smooth infinity section $D = \mathbf{P}_{\text{lines}}(A^{-1})$ with $D^2 = 0$ and $N_{D/S} \simeq A$.

These examples form large families. For fixed B of genus $g \geq 1$, a Poincaré line bundle gives an algebraic family of such total spaces over the g-dimensional abelian variety $\mathrm{Pic}^0(B)$. The torsion subgroup is countable, so the preceding conclusion holds for every parameter outside a countable set, hence for a very general member of this family. This is a statement within the indicated family, not a genericity assertion about all quasi-projective surfaces.

Moreover, let $\mu : \widetilde{X} \to X$ be the blowup of any finite set of n distinct points of X. Then $\widetilde{X}$ is again smooth and quasi-projective, and $\mu_* \mathcal{O}_{\widetilde{X}} = \mathcal{O}_X$ gives $\mathcal{O}(\widetilde{X}) = \mathbb{C}$. The pullback of the squared-norm exhaustion is smooth and psh, and remains proper because μ is proper. Blowing up the same points in S leaves the infinity section and its normal bundle unchanged. Since X retracts onto B, the rational Betti numbers of $\widetilde{X}$ are

$$b_1(\widetilde{X}) = 2g, \qquad b_2(\widetilde{X}) = n + 1.$$

Thus both Betti numbers can be arbitrarily large within this class.

Thus the constant-function case occurs even for smoothly weakly complete quasi-projective surfaces with a connected square-zero boundary. Quasi-projectivity supplies the divisor compactification used in our argument; it does not supply the nonconstant holomorphic function required by Proposition 5.1.

Corollary 5.9 (Continuous collars along square-zero curves)**.** *Let C be a smooth connected curve with $C^2 = 0$ in a smooth compact connected Kähler surface S. Suppose a neighborhood V of C admits a continuous plurisubharmonic function on $V \setminus C$ tending to $+\infty$ along C. Then $S \setminus C$ admits a smooth plurisubharmonic exhaustion. If $S \setminus C$ is not a modification of a Stein space, $\mathcal{O}_S(C)$ is unitary flat near C, and the divergent collar can be chosen pluriharmonic. No genus or Ueda-type hypothesis is required.*

Proof. Choose a smaller neighborhood of C compactly contained in V. The given function is bounded on a compact annulus at its outer boundary. Taking its maximum with a larger constant and extending by that constant gives a continuous plurisubharmonic exhaustion of $S \setminus C$. Apply Theorem 5.4. In its second alternative, the boundary consists of the single component C, so the positive kernel vector may be chosen to be 1, giving $A = C$. A smooth zero-curvature Hermitian metric is unitary flat. Equivalently, on simply connected coordinate neighborhoods, write the smooth pluriharmonic remainder in (5.16) as $h/c = 2\,\mathrm{Re}\, H$. Replacing a defining function w by $w \exp(-H)$ produces defining functions of equal modulus on overlaps, whose ratios are locally constant and unitary. □

Remark 5.10 (Scope and the weighted boundary). For a reducible boundary the flat metric is on the weighted line bundle $\mathcal{O}_S(A)$; flatness of $\mathcal{O}_S(D)$ is not asserted. Positivity of every coefficient of A is what makes the logarithmic function exhaust every end. The argument proves existence of a new smooth exhaustion and does not assert smooth approximation of the original continuous function. It covers all smooth quasi-projective surfaces and, more generally, divisor complements in compact Kähler surfaces. It does not provide a divisorial compactification of an arbitrary noncompact Kähler surface, or treat an arbitrary analytic domain in a projective surface whose boundary is not a divisor.

5.4. **Nonprojective compactifiers.**

Theorem 5.11 (Nonprojective boundary criterion)**.** *Let S be a smooth compact connected nonprojective surface and let $D \neq 0$ be effective. Then the following are equivalent:*

(i) *$S \setminus \operatorname{supp} D$ has a continuous psh exhaustion;*
(ii) *$S \setminus \operatorname{supp} D$ has a smooth psh exhaustion;*
(iii) *no connected component of $\operatorname{supp} D$ has negative definite intersection matrix.*

No Kähler or SNC hypothesis is required. The Hopf branch of the proof uses the elliptic-neighborhood theorem in the preprint [14, Theorem 1.6].

Proof. Only (iii)⇒(ii) needs proof, by Lemma 5.3(i). We separate the algebraic-dimension cases, using the classification facts in [8, Section 2].

If $a(S) = 1$, the algebraic reduction is a holomorphic elliptic fibration $f : S \to B$ with connected fibers, and every curve is vertical. For a fiber $F = \sum m_i C_i$, its matrix I satisfies

$$x^{\mathsf{T}} I x = -\sum_{i<j} (C_i \cdot C_j) m_i m_j \left(\frac{x_i}{m_i} - \frac{x_j}{m_j} \right)^2 . \tag{5.17}$$

The kernel is the line spanned by (m_i), and every proper principal submatrix is negative definite. Hence the boundary contains every fiber support it meets. Thus $X = f^{-1}(B \setminus P)$ for a finite nonempty set P, and a smooth strictly subharmonic exhaustion of the punctured compact curve pulls back to a smooth psh exhaustion of X. This argument includes reducible and multiple fibers.

Suppose $a(S) = 0$. For every line bundle L on a nonprojective surface, $c_1(L)^2 \leq 0$ [8, Lemma 2.1]. Every boundary matrix is therefore negative semidefinite. Hypothesis (iii) and the same Perron–Frobenius argument give an effective divisor A with positive integral coefficients, full boundary support, and $A^2 = 0$. Contract to a smooth minimal model $p : S \to S_0$. At each blowdown $q : T \to T'$, write $A = q^* A' + bE$, where $A' = q_* A$ and $E^2 = -1$. Then

$$0 = A^2 = (A')^2 - b^2, \qquad (A')^2 \leq 0,$$

so $b = 0$ and $A' \neq 0$. Consequently $A = p^*A_0$, $A_0^2 = 0$, and

$$\operatorname{supp} A = p^{-1}(\operatorname{supp} A_0).$$

The restriction $X \to S_0 \setminus \operatorname{supp} A_0$ is proper. It is enough to construct a smooth exhaustion downstairs.

In the Kähler case, S_0 is a torus or a K3 surface of algebraic dimension zero. A torus of algebraic dimension zero has no curves: an irreducible curve would be elliptic (by adjunction and the absence of rational curves), and its elliptic quotient would give positive algebraic dimension. On a K3 surface, Riemann–Roch gives $h^0(\mathcal{O}(A_0)) \geq 2$ because $A_0^2 = 0$ and $-A_0$ is not effective. Both alternatives contradict $a(S_0) = 0$.

In the non-Kähler case, S_0 is of class VII. If $b_2(S_0) = n > 0$, Enoki's Main Theorem [9, p. 454] identifies a nonzero effective square-zero divisor with a multiple of the distinguished cycle on an Enoki surface. Its complement is an affine-line bundle of degree $-n$ over an elliptic curve [9, (0.4), p. 453]. Complete the affine fibers to a projective $\mathbf{P}^1$-bundle. Its infinity section has square $n > 0$, so Lemma 5.3(iii),(ii) gives a smooth psh exhaustion of the affine-line complement. This uses Enoki's specific square-zero-divisor theorem, not a classification of all class VII surfaces.

If $b_2(S_0) = 0$, a minimal class VII surface containing a curve is elliptic or Hopf [9, p. 453]; algebraic dimension zero leaves Hopf. Each irreducible curve has arithmetic genus one, since $C^2 = K_{S_0} \cdot C = 0$. It cannot have rational normalization: a map from $\mathbf{P}^1$ lifts to $\mathbb{C}^2 \setminus \{0\}$ and is constant. Thus the curves are smooth elliptic and mutually disjoint. For each of them, finite Ueda type gives a smooth strictly psh divergent collar [13, Theorem 2.5]. Infinite type gives a unitary flat neighborhood by [14, Theorem 1.6], and hence the smooth pluriharmonic collar $-\log|w_i|^2$ in unitary defining functions. Lemma 5.3(ii) finishes the proof. □

In particular, for a nonprojective Kähler compactifier, a nonempty boundary complement can be continuously weakly complete only when $a(S) = 1$ and the boundary is a union of whole fiber supports. The $a(S) = 0$ Kähler branch is excluded, not an unresolved positive case.

5.5. **Smooth rulings.**

Theorem 5.12 (Boundary criterion for a smooth ruling)**.** *Let $f : S \to B$ be a smooth $\mathbf{P}^1$-bundle over a compact connected Riemann surface, and let $D \neq 0$ be reduced SNC. Then continuous psh exhaustion, smooth psh exhaustion, and the absence of a negative definite connected boundary component are equivalent for $S \setminus D$. There is no restriction on the degree $D \cdot F$.*

Proof. Necessity follows from Lemma 5.3. For sufficiency, positive largest eigenvalue gives a smooth collar. A holomorphic $\mathbf{P}^1$-bundle over a compact curve is a projectivized rank-two bundle; in particular, it is projective and has a section. For a connected component $E = \bigcup E_i$ with zero largest eigenvalue,

take $A = \sum a_i E_i$ as in (5.1). We construct a smooth exhaustion of $S \setminus E$, whose restriction supplies the required collar at E.

If every component is vertical, E is one whole fiber, since the fibers of a smooth ruling are irreducible. Pull back an exhaustion of B minus its image. Otherwise $A \cdot F > 0$, so no fiber can be a component of E. Choose a section C_0, with $C_0^2 = -e$; the numerical divisor space has basis C_0, F. Writing $A \equiv aC_0 + bF$, the equality $A^2 = 0$ gives $b = ae/2$. Writing $E_i \equiv d_i C_0 + b_i F$, the identities $A \cdot E_i = 0$ give

$$E_i \equiv d_i(C_0 + (e/2)F), \qquad E_i^2 = E_i \cdot E_j = 0 \quad (i \neq j).$$

Distinct components are therefore disjoint. Connectedness and the SNC hypothesis make $E = C$ a single smooth square-zero curve. For $d = C \cdot F$, adjunction and $K_S \equiv -2C_0 + (2g(B) - 2 - e)F$ give

$$2g(C) - 2 = d(2g(B) - 2).$$

Riemann–Hurwitz shows that $C \to B$ is finite étale.

If $d = 1$, the complement is an affine-line bundle. Its linear bundle is dual to $N_{C/S}$ and therefore has degree zero. We recall the smooth exhaustion construction of Koike–Ueda [15, Proposition 2.1]. In unitary flat frames its transitions are $w_i = a_{ij} w_j + b_{ij}(z)$, $|a_{ij}| = 1$. Choose a smooth section and subtract a global smooth section of the linear bundle so that its $\bar\partial$ is harmonic. On coordinate discs, choose antiholomorphic primitives g_i of this harmonic form. Changing the affine coordinates by holomorphic translations then gives

$$w_i = a_{ij} w_j + c_{ij}, \qquad g_i = a_{ij} g_j + c_{ij},$$

where c_{ij} is constant: the differences of the g_i are both holomorphic and antiholomorphic. The functions $\rho_i = |w_i - g_i|^2$ agree and have Levi matrix

$$\begin{pmatrix} |(g_i)_{\bar z}|^2 & 0 \\ 0 & 1 \end{pmatrix}.$$

They define a smooth psh exhaustion; its sublevels are bounded closed disc bundles around a smooth section over compact B.

For $d \geq 2$, a finite étale Galois base change splits the boundary into d disjoint sections. Each has square zero, since the pullback of C is their disjoint sum and the projection formula applies. If $d = 2$, putting these sections at 0 and ∞ identifies the complement with $L^\times$ for a degree-zero line bundle. A unitary flat metric makes $(\log \|v\|^2)^2$ a smooth psh exhaustion. If $d \geq 3$, three labeled sections trivialize the $\mathbf{P}^1$-bundle; the others are constant, as maps from compact B' to $\mathbf{P}^1$ avoiding $0, 1, \infty$. The complement is $B' \times (\mathbf{P}^1 \setminus Q)$ for a finite set Q, and a smooth subharmonic exhaustion of the punctured projective line pulls back. Average a nonnegative exhaustion over the finite deck group. The average remains proper, being at least the original exhaustion divided by the group order, and descends smoothly to $S \setminus C$. Finally glue the collars at all connected boundary components using Lemma 5.3(ii). □

This numerical existence criterion uses the actual smooth ruling, not merely birational ruledness. Regularization for arbitrary smooth projective blowups and singular rational fibrations is nevertheless covered by Theorem 5.4; the numerical criterion alone is not asserted for those compactifiers. For $S = \mathbf{P}^2$, every nonzero effective divisor is ample, so its complement has a smooth psh exhaustion directly from a positive metric on $\mathcal{O}_S(D)$.

5.6. **Direct geometric constructions in the projective case.** The conditional regularization conclusions in the next proposition are now special cases of Theorem 5.4. We retain their proofs because they give direct geometric constructions from disjoint boundary divisors, moving divisors, surface abundance, and fibrations.

Proposition 5.13 (Projective affirmative cases)**.** *Let S be smooth projective, let $D \neq 0$ be reduced SNC, and assume $X = S \setminus D$ has a continuous psh exhaustion. Then X has a smooth psh exhaustion in each of the following cases:*

(i) *D is disconnected;*
(ii) *on every connected boundary component with zero largest intersection eigenvalue, a positive integral kernel divisor A satisfies $\kappa(S, A) = 1$ or $K_S \cdot A < 0$;*
(iii) *K_S is torsion in $\mathrm{Pic}(S)$;*
(iv) *every component of D is vertical for a morphism $f : S \to B$ with connected fibers onto a smooth projective curve.*

Proof. For (i), write the connected boundary components as $E_1, \dots, E_r$, $r \geq 2$. The matrix argument provides effective A_k with support E_k and $A_k^2 \geq 0$, with strict inequality in the positive-eigenvalue case. Since $A_k \cdot A_l = 0$ for $k \neq l$, the Hodge index theorem excludes a positive square and makes the nonzero square-zero nef classes proportional. The proportionality factors are positive rational numbers, by intersection with an integral ample class. Choose positive integers p_k, q_k such that $p_k A_k - q_k A_1$ is numerically trivial, for $k \geq 2$. Its real first Chern class is zero by Hodge index, and the $\partial\bar{\partial}$-lemma gives a unitary flat metric h_k on $\mathcal{O}_S(p_k A_k - q_k A_1)$. For its canonical meromorphic section σ_k, the functions

$$\phi_k = \log \|\sigma_k\|_{h_k}^2, \qquad \rho = \sum_{k=2}^{r} \phi_k^2$$

are smooth on X; the ϕ_k are pluriharmonic, so ρ is psh. It tends to infinity at every boundary component, proving properness.

For (ii), a nef divisor of Iitaka dimension one on a surface is semiample [11, Lemma 7.3]. If $K_S \cdot A < 0$, Riemann–Roch gives

$$\chi(\mathcal{O}_S(nA)) = \chi(\mathcal{O}_S) - \tfrac{n}{2} K_S \cdot A \longrightarrow +\infty.$$

Serre duality and intersection with an ample class give $H^2(S, \mathcal{O}_S(nA)) = 0$ for large n. Thus $\kappa(S, A) \geq 1$, and nefness with $A^2 = 0$ gives $\kappa(S, A) = 1$.

In either case A is semiample. A base-point-free multiple defines, after Stein factorization, a morphism $f : S \to B$ to a smooth curve with connected fibers and $\mathcal{O}_S(mA) = f^*L$. The canonical section descends because $f_*\mathcal{O}_S = \mathcal{O}_B$, so $mA = f^*G$ for a nonzero effective divisor G on B. Hence

$$S \setminus \operatorname{supp} A = f^{-1}(B \setminus \operatorname{supp} G),$$

with proper restricted map. Pulling back an exhaustion of the punctured curve gives a collar at $\operatorname{supp} A$. Combine these with the positive-eigenvalue collars using Lemma 5.3(ii).

For (iii), consider such a square-zero nef A. Choose a small positive rational ε so that every coefficient of εA is at most one. The log divisor $K_S + \varepsilon A$ is nef. Log abundance for surfaces [10, Theorem 6.1] makes it semiample. Killing the torsion of K_S and clearing denominators shows that A itself is semiample. Apply the preceding fibration argument. This is a direct consequence of log abundance on the given pair; no change of the open surface by a log minimal model program is involved.

For (iv), identity (5.17) holds for the multiplicity vector of any fiber. Every proper subset of its components has negative definite intersection matrix. Therefore the boundary obstruction forces D to be a union of whole fiber supports. The restricted map $X \to B \setminus P$ is proper for a finite nonempty set P; pull back an exhaustion of the punctured base. □

In (i) no SNC assumption is necessary, because boundary resolution preserves the complement and its number of boundary components. Part (iii) applies, in particular, to the minimal projective surfaces of Kodaira dimension zero: K3, Enriques, abelian, and bielliptic surfaces [7, Chapter VI]. This is a standard consequence of the cited abundance theorem, not a new classification result. Part (iv) includes the vertical-boundary case of an elliptic fibration, while arbitrary horizontal boundaries under the continuous-exhaustion hypothesis are covered by Theorem 5.4.

The weighted square-zero configurations that arise from the boundary matrix are all covered by Theorem 5.4 under the continuous-exhaustion hypothesis. They therefore do not constitute a remaining projective counterexample class.

5.7. **A concise smooth-boundary criterion.** The following sufficient conditions remain useful because no continuous exhaustion is assumed. If a continuous collar is already given, the unrestricted square-zero regularization statement is Corollary 5.9.

For a smooth square-zero curve C, its normal bundle N is unitary flat. The Ueda type measures the orders to which local defining functions can be made to have unitary linear transition functions. Infinite type is formal; a unitary flat neighborhood requires *convergent* defining functions $w_i = t_{ij}w_j$, $|t_{ij}| = 1$. These two conditions must not be conflated [12, 13].

Proposition 5.14 (Sufficient conditions, not a classification)**.** *Let S be a compact Kähler surface and let $C \subset S$ be smooth, connected, and square-zero.*

Write $\rho : H_1(C,\mathbb{Z}) \to \mathrm{U}(1)$ *for the unitary character of* $N_{C/S}$ *and* $K = \ker[H_1(C,\mathbb{Z}) \to H_1(S,\mathbb{Z})]$. *Then* $S \setminus C$ *has a smooth psh exhaustion if any of the following holds: finite Ueda type; a unitary flat neighborhood; torsion or Diophantine normal bundle;* $\rho(K)$ *finite;* $q(S) \geq g(C)$*; or* $g(C) = 1$. *No continuous exhaustion is assumed. Only the last case uses* [14, Theorem 1.6].

Proof. Finite type gives a smooth strictly psh collar [13, Theorem 2.5]. A flat neighborhood gives the pluriharmonic collar $-\log|w_i|^2$. In infinite type, torsion or Diophantine N gives such a neighborhood by Ueda's convergence theorem [12, Theorem 3]; finite type has already been handled. Here Diophantine means $\mathrm{dist}(N^m, \mathcal{O}_C) \geq cm^{-\tau}$ for some $c, \tau > 0$ and all $m \geq 1$.

If $\rho(K)$ is finite, a power ρ^k is trivial on K, hence defines a character of the image of $H_1(C,\mathbb{Z})$ in $H_1(S,\mathbb{Z})$. It extends to the whole group because $\mathrm{U}(1)$ is divisible. In the remaining non-torsion, infinite-type case, Neeman's extension theorem in [13, Theorem 4.4] gives an effective divisor E disjoint from C and cohomologous to mC. A unitary flat metric on $\mathcal{O}_S(mC - E)$ then gives the pluriharmonic collar $-m^{-1}\log\|\sigma\|^2$ at C.

For $q(S) \geq g(C)$, consider $r : H^1(S,\mathcal{O}_S) \to H^1(C,\mathcal{O}_C)$. If r is not injective, [13, Proposition 2.7] makes a multiple of C a fiber of a fibration, and the punctured-base exhaustion applies. Otherwise the dimension inequality makes r an isomorphism. Hodge decomposition and duality make $H_1(C,\mathbb{Q}) \to H_1(S,\mathbb{Q})$ an isomorphism. Its integral kernel is torsion and lies in the free abelian group $H_1(C,\mathbb{Z})$, so $K = 0$. Apply the character case. Finally, when $g(C) = 1$, infinite type implies a unitary flat neighborhood by [14, Theorem 1.6]. In every case finish with Lemma 5.3(ii). □

The finite-type and flat-neighborhood arguments also apply without a Kähler hypothesis, as used in the Hopf branch above. Neither semipositivity of $\mathcal{O}_S(C)$ nor holomorphic convexity of $S \setminus C$ is asserted as a necessary condition: the proposition constructs collars, not a general equivalence with either of those properties.

5.8. **Classification of compactifiers and the scope of the theorem.** In Table 1, S is a smooth compact connected surface, $D \neq 0$ is a divisor, and $X = S \setminus \operatorname{supp} D$ is assumed to have a continuous psh exhaustion. The familiar minimal types are listed to organize the compactifiers; the projective conclusion holds for every smooth projective model, including nonminimal ones. "Affirmative" means that a smooth psh exhaustion exists; "excluded" means that no such nonempty divisor complement satisfies the continuous-exhaustion hypothesis. The table uses the Enriques–Kodaira classification [7, Chapter VI], and the individual exhaustion conclusions follow from the theorems above.

All projective Kodaira dimensions are covered by Theorem 5.4. In particular, rational blowups, horizontal boundaries on properly elliptic surfaces, and connected square-zero boundaries on surfaces of general type require no additional hypothesis. The proof works on the given compactifier; it does

not contract curves meeting the open surface. The nonprojective existence criterion likewise includes nonminimal compactifiers through its full-pullback argument.

The high-genus, infinite-Ueda-type collar problem is covered by Corollary 5.9. When the complement is not a modification of Stein, the current argument gives a smooth flat metric near the boundary; otherwise the Stein modification gives the smooth exhaustion. Neither infinite Ueda type alone nor the numerical square-zero condition is asserted to imply existence of a continuous exhaustion.

The unrestricted continuous-to-C^2 problem for complex surfaces, including arbitrary noncompact Kähler surfaces without a divisor compactification, is not resolved here. Nor does the theorem treat every analytic domain in a projective surface: a real or otherwise nondivisorial boundary is outside its hypotheses. The new compactifiable Kähler regularization theorem is independent of the 2026 elliptic-neighborhood input used in the separate existence criteria, and independent of the threefold constructions.

TABLE 1. Divisor complements by compactifier type; a continuous psh exhaustion is assumed. Projective conclusions include nonminimal models.

Compactifier S	Exhaustion conclusion and exact scope
Projective, $\kappa(S) = -\infty$: rational or ruled	**Affirmative.** Every divisor complement satisfying the hypothesis, including arbitrary smooth projective blowups; Theorem 5.4.
Projective, $\kappa(S) = 0$: K3, Enriques, abelian, bielliptic models	**Affirmative.** All boundaries and nonminimal models; Theorem 5.4. The torsion-canonical construction remains a direct alternative on the corresponding minimal models.
Projective, $\kappa(S) = 1$: properly elliptic	**Affirmative.** Horizontal as well as vertical boundaries; Theorem 5.4.
Projective, $\kappa(S) = 2$: general type	**Affirmative.** All boundaries, including connected weighted square-zero divisors; Theorem 5.4.
Nonprojective, $a(S) = 1$	**Affirmative.** The boundary is a union of whole elliptic fiber supports; Theorem 5.11. This includes the relevant Kodaira and elliptic surfaces.
Nonprojective Kähler, $a(S) = 0$: torus or K3 models	**Excluded.** No nonempty divisor complement satisfies the hypothesis; see the proof of Theorem 5.11.
Nonprojective, $a(S) = 0$, class VII, $b_2 = 0$ on the minimal model	**Affirmative when a boundary exists.** The minimal model is Hopf; the elliptic-collar argument uses [14]. Blowups of curve-free minimal models have only exceptional boundaries, excluded by negative definiteness.
Nonprojective, $a(S) = 0$, class VII, $b_2 > 0$ on the minimal model	**Affirmative for every admissible boundary.** The matrix condition forces the Enoki square-zero-divisor case; other boundaries are excluded by the negative definite obstruction.

6. Holomorphic convexity and regularization for curve fiberings

This section gives a regularization criterion independent of the compactification hypotheses in Section 5. Complex spaces are reduced, Hausdorff, and have countable topology. Pseudoconvexity means the existence of a continuous real-valued psh exhaustion, as in [20, p. 455]. All fiber dimensions and irreducible components refer to the reduced supports of the fibers.

Theorem 6.1. *Let X be a complex space of pure dimension $n \geq 2$, admitting a continuous real-valued psh exhaustion ρ. Suppose $F : X \to \mathbb{C}^{n-1}$ is holomorphic and every nonempty fiber has dimension at most one. Then X is holomorphically convex and admits a smooth psh exhaustion. No condition on* $\operatorname{Sing} X$ *is required.*

Neither normality nor Cohen–Macaulayness of X is assumed. The map F may be nonproper and its fibers may be disconnected, reducible, singular, or multiple. The proof concerns arbitrary singularities of the total space as well as singular fibers. Throughout the proof, X, ρ, F satisfy the hypotheses of the theorem.

6.1. **Cycle-theoretic and convexity inputs.** We state the exact forms needed.

(C1) *Geometric flatness over a normal base.* An equidimensional holomorphic map of reduced pure-dimensional complex spaces to a normal complex space has an analytic family of effective weighted fiber cycles, with support equal to the corresponding fiber. Cycles may be noncompact and locally finite; the map need not be proper. See [21, Chapter I, §1(b), p. 816], with the definitions on pp. 814–815. See also [23, Definition 1.1.1 and following Remark (ii); author preprint pp. 3–4], referring to [22, Chapter IV, Theorem 3.4.1]. This is a statement about cycles, not flatness of the local rings.

(C2) *Continuity of integration.* For a continuous family Z_t of analytic d-cycles on a complex space and a continuous compactly supported (d,d)-form ω in the local-embedding sense, $t \mapsto \int_{Z_t} \omega$ is continuous. See [22, Chapter IV, Proposition 2.3.1]; this exact location is also cited in [24, proof of Theorem 4.0.7, Step 4, author preprint p. 21]. Compact support on the ambient space supplies the required proper support on the graph of the family.

(C3) *Bishop compactness on a complex space.* For any complex analytic space W, compact effective d-cycles whose supports lie in a fixed compact set and whose Hermitian volumes are uniformly bounded form a relatively compact subset of the compact-cycle space. See [25, Theorem 1, p. 26]. The topology on pp. 25–26 has the following properties: under convergence, containment of the limiting support in an open set implies eventual containment of the approximating

supports there; local integration currents also converge. The cycles need not meet W_{reg}.

The following results are used in their stated complex-space forms in [20].

(V1) *Psh sublevel criterion* (Proposition 2.2, p. 456). For a continuous real-valued psh function u on a complex space W, the space W is holomorphically convex if and only if $\{u < t\}$ is holomorphically convex for every real t.

(V2) *Convexity over a Stein base* (Proposition 2.4, p. 457). A pseudoconvex space holomorphically convex over a Stein space is holomorphically convex. Here convexity over the base means that each base point has an open neighborhood with holomorphically convex inverse image.

(V3) *Boundary-collar criterion* (Corollary 2.5, p. 457). Let $\pi : M \to S$ be holomorphic, with S Stein, and let $D \subset M$ be pseudoconvex and open. If $\overline{D}^{M} \to S$ is proper and an open neighborhood U of $\partial_M D$ has Stein reduced fibers under π, then D is holomorphically convex.

Remark 6.2 (Exhaustion clarification in V3). The printed proof of Corollary 2.5 calls a smooth auxiliary function constructed across $\partial_M D$ a relative exhaustion. One must add the psh exhaustion of D. After shrinking to a relatively compact base neighborhood V, with closure in a larger such neighborhood, that construction gives a smooth h, defined near $\overline{D} \cap \pi^{-1}(\overline{V})$, strictly psh near its relative boundary. This set is compact, so h is bounded there. For a continuous psh exhaustion η_D of D, the function $\eta_D + h$ on $D \cap \pi^{-1}(V)$ is a vertical exhaustion. Outside the boundary collar, the relevant part of $\overline{D} \cap \pi^{-1}(\overline{V})$ is a compact subset of D; the sum is bounded above there. Above this bound it is strictly psh. This satisfies the definition of a 1-convex map on pp. 456–457. Theorem 2.3 and Proposition 2.4 of that paper complete the argument. No smoothness of M or its fibers is required for this completion.

We also use that a reduced complex curve without compact irreducible components is Stein; see [4, §3, p. 1663]. The empty space is included as Stein.

6.2. **Fiber cycles and stability.**

Lemma 6.3 (Uniform area of compact fiber components)**.** *Fix a Hermitian metric h on X, defined using local embeddings. For every compact $L \subset X$ and $z_0 \in \mathbb{C}^{n-1}$, there are a neighborhood B of z_0 and $A > 0$ such that $\int_C h \le A$ for every reduced compact irreducible fiber component $C \subset L$ over B.*

Proof. On each local irreducible branch of X, a fiber is cut out by $n-1$ holomorphic functions. The analytic dimension theorem (the generalized principal ideal theorem for local analytic rings) gives dimension at least $n - (n-1) = 1$ for every component of this zero set. The upper bound therefore makes every nonempty reduced fiber pure one-dimensional. This

also holds for the restriction to each irreducible source component, as required by C1.

Since the base is normal, C1 supplies analytic weighted fiber cycles $\mathcal{F}_z$ with $|\mathcal{F}_z| = F^{-1}(z)$. Every component has a positive integer coefficient. Choose a nonnegative smooth compactly supported function χ equal to one near L. By C2, the function

$$z \longmapsto \int_{\mathcal{F}_z} \chi h$$

is continuous and hence bounded by some A near z_0. Positivity gives

$$\int_C h = \int_{[C]} \chi h \le \int_{\mathcal{F}_z} \chi h \le A.$$

There is no need to identify these geometric multiplicities with scheme-theoretic fiber multiplicities. □

Lemma 6.4 (Stability of noncompact components)**.** *Let $K \subset X$ be compact. If every irreducible component of $F^{-1}(z_0)$ meeting K is noncompact, then the same holds for every fiber over some neighborhood of z_0.*

Proof. The assertion is immediate if K is empty. If the central fiber misses K, a ball avoiding the compact set $F(K)$ suffices. Otherwise suppose the assertion fails. There are $z_j \to z_0$, compact irreducible components $C_j \subset F^{-1}(z_j)$, and $x_j \in C_j \cap K$. Pass to a subsequence with $x_j \to x \in K$. The function ρ is constant on each C_j: its pullback to the compact connected Riemann surface normalizing C_j is subharmonic, hence constant. Thus

$$C_j \subset L := \{\rho \le \max_K \rho\},$$

a fixed compact set. Lemma 6.3 bounds their areas, so C3 gives a subsequence $[C_j]$ converging to a compact effective one-cycle T.

The limit is nonzero: otherwise upper support-continuity applied to the empty open set forces C_j eventually to be empty. Moreover $x \in |T|$. If not, choose an open neighborhood of the compact $|T|$ whose closure misses x. It eventually contains every C_j, contradicting $x_j \to x$.

Also $|T| \subset F^{-1}(z_0)$. A point outside the central fiber has a neighborhood on which F avoids a neighborhood of z_0. This neighborhood misses every sufficiently large C_j; local convergence of currents shows that T has no support there. An irreducible component of $|T|$ through x is now a compact closed irreducible analytic curve in the pure one-dimensional central fiber. A closed irreducible analytic subset of the same dimension as an irreducible curve containing it equals that curve. It is therefore a full irreducible component of the central fiber, contradicting the hypothesis. □

Remark 6.5. This replaces the smooth-source Corollary 4.2 of [20, p. 462]. The curves C_j, or their limit, may lie entirely in $\operatorname{Sing} X$. Multiple fibers are allowed because effective weighted fiber cycles dominate their reduced component cycles.

6.3. **Convex neighborhoods and globalization.**

Lemma 6.6 (Convex sublevel neighborhoods)**.** *For every $\lambda \in \mathbb{R}$ and $z_0 \in \mathbb{C}^{n-1}$, there are $c > \lambda$ and a ball $B = B_r(z_0)$ such that $D = F^{-1}(B) \cap \{\rho < c\}$ is holomorphically convex. Consequently $F^{-1}(B) \cap \{\rho < \lambda\}$ is holomorphically convex.*

Proof. Let S be the set of constant ρ-values on the compact irreducible components of $F^{-1}(z_0)$. For each real R, only finitely many such components have value at most R: they meet the compact set $\{\rho \le R\}$, while the irreducible components of a fixed analytic set are locally finite. Thus S is locally finite. Choose

$$\lambda < a < b, \qquad [a,b] \cap S = \varnothing, \qquad K = \{a \le \rho \le b\}.$$

Every central irreducible component meeting K is noncompact. Lemma 6.4 gives a ball $B = B_r(z_0)$ with the same property for all fibers over B. Choose $c \in (a,b)$ and put

$$M = F^{-1}(B), \qquad D = M \cap \{\rho < c\}, \qquad U = M \cap \{a < \rho < b\}.$$

Then $\partial_M D \subset U$.

No reduced curve $U \cap F^{-1}(z)$ has a compact irreducible component A. Such an A, being compact and analytic in U, would be a closed analytic curve in X. It would therefore be a full irreducible component of the pure one-dimensional full fiber, meeting K, a contradiction. Thus the Stein-curve criterion makes all reduced U-fibers Stein.

A continuous psh exhaustion of D is

$$\eta(x) = \max\left\{ \frac{1}{c - \rho(x)}, \frac{1}{r^2 - \|F(x) - z_0\|^2} \right\}.$$

Both terms are psh by convex increasing composition. For $t > 0$, its closed t-sublevel lies in

$$\{\rho \le c - 1/t\} \cap F^{-1}\{\|z - z_0\|^2 \le r^2 - 1/t\},$$

is closed there, and stays inside D; hence it is compact. For $t \le 0$ it is empty. Furthermore, $\overline{D}^M \to B$ is proper: over a compact $T \subset B$ its inverse image is a closed subset of the compact set $\{\rho \le c\} \cap F^{-1}(T)$. The boundary-collar criterion V3 proves that D is holomorphically convex. The final assertion follows from V1 for $\rho|_D$. □

Proof of Theorem 6.1. For $X(\lambda) = \{\rho < \lambda\}$, Lemma 6.6 says that $F|_{X(\lambda)}$ is a holomorphically convex map to $\mathbb{C}^{n-1}$. The space $X(\lambda)$ is pseudoconvex with exhaustion $1/(\lambda - \rho)$. Criterion V2 therefore makes it holomorphically convex. This holds for every λ, so V1 proves that X is holomorphically convex.

Let $q : X \to Z$ be its proper Remmert reduction, with Z Stein. We use the standard implication from holomorphic convexity to smooth weak completeness, recalled in [4, p. 1659]. Choose a smooth strictly psh exhaustion

u of the Stein complex space Z. As in Proposition 5.1, the pullback $u \circ q$ is smooth in the local-embedding sense and psh. It is exhaustive because

$$\{u \circ q \leq t\} = q^{-1}(\{u \leq t\})$$

is compact for every real t. This proves the smooth-exhaustion assertion and also Theorem 1.4. □

Corollary 6.7 (Simultaneous contraction)**.** *There are a Stein space Z and a proper surjective holomorphic map $q : X \to Z$, with connected fibers and $q_*\mathcal{O}_X = \mathcal{O}_Z$, whose nontrivial fibers are the maximal connected unions of compact irreducible fiber curves of F. The map F factors holomorphically through q.*

Proof. Take the Remmert reduction of X, using the reduction theory recalled in [20, proof of Proposition 2.4, p. 457]. The sheaf property makes the components of F descend, so each q-fiber lies in an F-fiber and has dimension at most one. A positive-dimensional compact connected analytic set of dimension at most one is a finite connected union of compact irreducible curves. Each is a full component of the corresponding pure one-dimensional F-fiber. Conversely, every compact irreducible curve is mapped to a point in the Stein space Z; hence so is any connected chain of them. Such a chain has constant ρ and lies in a compact sublevel. Local finiteness of the components of its F-fiber therefore makes each maximal connected union finite. This identifies the fibers of q. □

Remark 6.8. For arbitrary reduced X, normality of Z is not asserted; it holds if X is normal. Global holomorphic convexity supplies the contraction, without requiring an independent gluing of locally defined quotients.

Remark 6.9 (Scope of the regularization conclusion). The theorem constructs a smooth psh exhaustion; it does not approximate the prescribed continuous exhaustion or assert that the pullback from the Remmert reduction is strictly psh. Pure dimension and continuity of the real-valued exhaustion are used in the proof. The broader convention allowing an upper semicontinuous, possibly $-\infty$-valued exhaustion in [4] is not treated here.

In particular, a continuously weakly complete pure n-dimensional space without a smooth psh exhaustion cannot admit a holomorphic map to $\mathbb{C}^{n-1}$ all of whose fibers have dimension at most one. For the threefolds in Theorems 1.1 and 1.2, this is also explained directly by their compact surfaces: the zero section in M and exceptional surfaces in N are mapped to points by every holomorphic map to $\mathbb{C}^2$.

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